PHYSICS

SEC-1 : (Maximum Marks: 40)

- This section contains **TEN (10)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

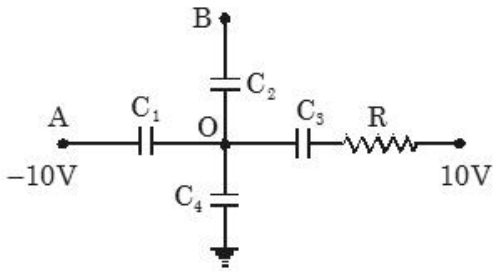
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

Space for Rough Work

1. Figure shows small part of circuit. [Given $C_1 = C_2 = C_3 = C_4 = 5\ \mu F$] Which of the following statement is/are true ?



(A) Potential of point O can be determined but that of B can't be determined.

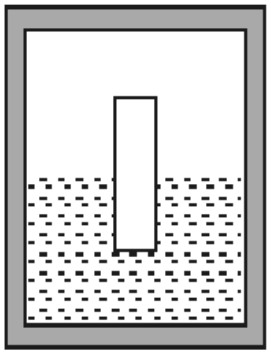
(B) Potential of both point O & B can't be determined.

(C) If charge on capacitor C_2 were also specified, potential of point O can be determined but that of B cann't be determined.

(D) If charge on capacitor C_2 were also specified, potential of both the point O & B can be determined.

Space for Rough Work

2. A cylinder having length 10cm, and density 0.65 g/cm^3 floats half submerged in the liquid shown. ($\rho_{air} = 1.3 \times 10^{-3}$ g/cm^3) Now pressure of air made 100 time of initial pressure keeping temperature constant. Which of the following statement(s) is/are true :



(A) The cylinder moves upwards.

(B) The cylinder moves downwards.

(C) Displacement of the cylinder is 0.56 cm.

(D) Displacement of the cylinder is 0.60 cm.

3. Assume that the earth changes its shape and turns into an infinite cylinder whose radius and distance of moon from the axis of cylindrical earth remains unchanged and also moon is remain spherical. Assume density of earth remains same and distance of moon from earth is much greater than radius of earth. Which of the following statement(s) is/are correct :

(A) Gravitational field at the moon due to earth increases.

(B) Gravitational field at the moon due to earth decreases.

(C) Speed of the moon in its orbit around the cylindrical earth increases.

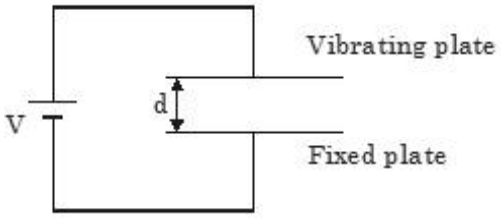
(D) Speed of the moon in its orbit around the cylindrical earth decreases.

Space for Rough Work

4. A particle of mass m is thrown in a parabolic path towards earth having mass M and radius R. It comes to a greatest approach distance R/2 from surface of earth. Magnitude of total energy change required to make the particle moving in an elliptical path having eccentricity 1/2, with the greatest approach point either aphelion or perihelion are :

(A) $\frac{1}{3}\frac{GMm}{R}$ (B) $\frac{1}{6}\frac{GMm}{R}$ (C) $\frac{1}{2}\frac{GMm}{R}$ (D) $\frac{GMm}{R}$

5. An air-filled parallel plate capacitor with the plate area A is connected to a battery with an emf V volt and negligible internal resistance. One of the plates vibrates so that the distance between plates varies as $d = d_0 + a\cos\omega t$ (where $a << d_0$). The capacitor breakes down when the instantaneous current reaches to the value of I. Choose the correct options(s).



(A) The maximum possible amplitude vibrations 'a' is $\frac{Id_0^2}{VA\omega\varepsilon_0}$

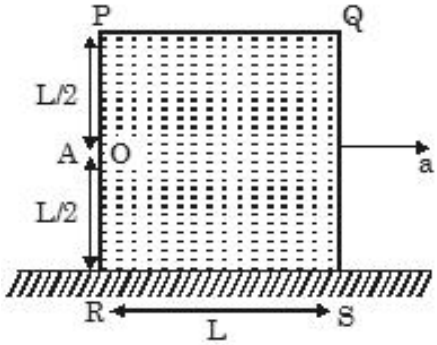
(B) The maximum possible amplitude vibrations 'a' is $\frac{2Id_0^2}{VA\omega\varepsilon_0}$

(C) The instantaneous current in circuit is $\frac{\varepsilon_0 V a}{(d_0 + a\cos\omega t)^2}\sin\omega t$.

(D) The instantaneous current in circuit is $\frac{\varepsilon_0 V a}{2(d_0 + a\cos\omega t)^2}\sin\omega t$.

Space for Rough Work

6. A small solid ball of density ρ is held inside at point A of cubical container of side L, filled with an ideal liquid of density 4ρ as shown in the figure. Now, if the container starts moving with constant acceleration a horizontally and the ball is released from point A simultaneously, then



(A) For ball to hit the top of container at end Q, a = 3g

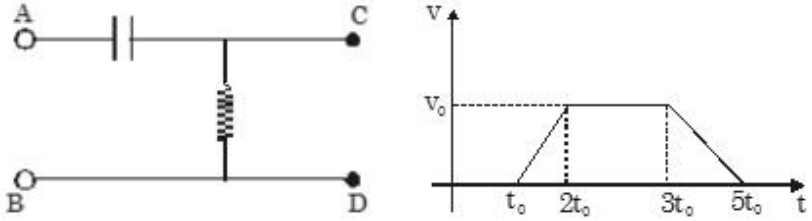
(B) For ball to hit the top of container at end Q, a = 2g

(C) Ball hits the top of container at end Q after a time $t = \sqrt{\frac{L}{3g}}$

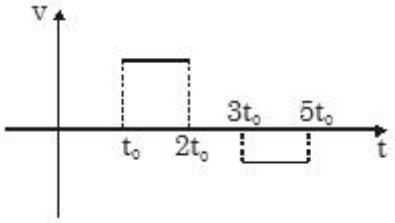
(D) Ball hits the top of container at end Q after a time $t = \sqrt{\frac{2L}{3g}}$

Space for Rough Work

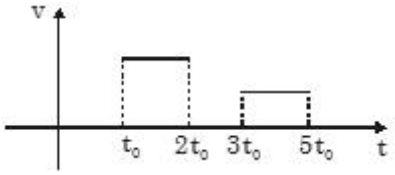
7. A time varying voltage is applied across AB as shown. Potential difference across the capacitor plate varies as shown. Then which of the following is/are correct.



(A) Plot of potential difference across CD is



(B) Plot of potential difference across CD is

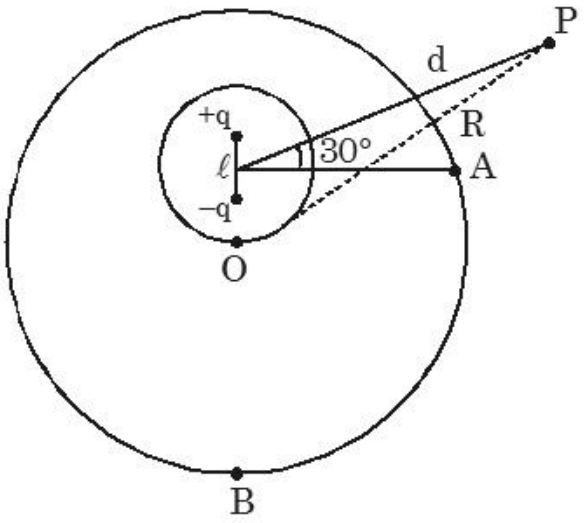


(C) Plot of current across resistance does not vary for large value of resistance & capacitor

(D) Plot of current across resistance does not vary for small value of resistance & capacitor

Space for Rough Work

8. A cavity is taken out from a uniform conducting sphere. Inside the cavity a dipole is placed as shown in the figure. Find which of the following is/are true (Given : $q = 10^{-1}$C, ℓ = 0.1 mm, θ = 30°, d = 10 cm, R = 12 cm)



(A) Potential at points A & B are same

(B) Potential at points A & B are different

(C) Potential at point P is 0 volt

(D) Potential at point P is 2 volt

Space for Rough Work

9. Water jet coming out of a stationary horizontal tube at speed v = 2m/s strikes horizontally a massive wall moving in opposite direction with same speed. Water comes to rest relative to wall after striking. Cross-section of jet is 3 cm^2 and density of water is 1 gm/cc. Choose the correct statement(s) :-

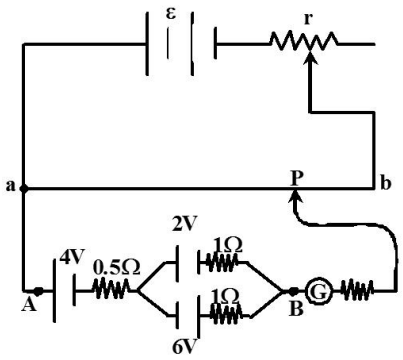
(A) Force exerted on the wall is 1.2 N

(B) Force exerted on the wall is 2.4 N

(C) Rate of change of kinetic energy of water jet striking the wall is 19.2 J/s.

(D) Rate of change of kinetic energy of water jet striking the wall is zero.

10. A potentiometer as shown in figure in which three batteries between A & B are connected when the point P is the null point, measured by the potentiometer arrangement. Then choose correct statement(s)



(A) Current passing through 6 V battery is 4A

(B) Current passing through 2 V battery is 4A

(C) $V_A - V_B = 2V$

(D) $V_A - V_B = 6V$

Space for Rough Work

SEC-2 : (Maximum Marks: 24)

This section contains **SIX (06)** questions.
This section contains **THREE** paragraphs.
Based on each paragraph, there are **TWO** questions
Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
For each question, choose the option(s) corresponding to (all) the correct answer(s)
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –2 In all other cases.
For Example : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

Space for Rough Work

Paragraph for Questions 11 and 12

When a conductor is charged then its potential rises. The increase in potential is directly proportional to the charge given to the conductor.

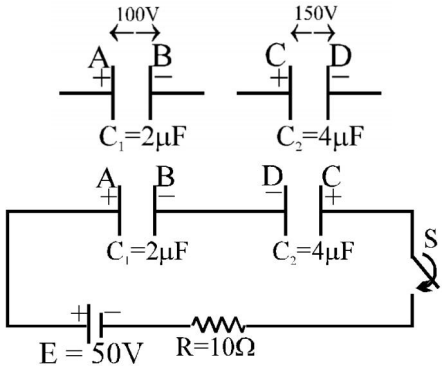
$Q \propto V \Rightarrow Q = CV$

The constant C is known as the capacity of the conductor.

Let C is capacitance of a conductor. On being connected to a battery. It charges to a potential V from zero potential. If q is charge on the conductor at that time then q = CV. Let battery supplies small amount of charge dq to the conductor at constant potential V. Then small amount of work done by the battery against the force exerted by exsiting charge is

$$dW = V\,dq = \frac{q}{C}dq \Rightarrow W = \int_0^Q \frac{q}{C}\,dq = \frac{1}{C}\left[\frac{q^2}{2}\right]_0^Q \Rightarrow W = \frac{Q^2}{2C}$$

Two capacitors of capacitance $C_1 = 2\mu F$ & $C_2 = 4\mu F$ are charged to Potential difference 100 V & 150V respectively. The charged capacitors are connected as shown in figure through a battery & resistance. At time t = 0 switch S is closed.



1. Choose the correct statement(s) :

(A) Current through the battery immediately after closing switch is 10 A.

(B) Current through the battery immediately after closing switch is 3 A.

(C) work done by battery is $\frac{20}{3}$ mJ

(D) work done by battery is $\frac{10}{3}$ mJ

Space for Rough Work

2. Final charges on each capacitor will be :

(A) $\frac{2000}{3}\ \mu C, \frac{1400}{3}\ \mu C$

(B) $\frac{1000}{3}\ \mu C, \frac{1800}{3}\ \mu C$

(C) $\frac{1200}{3}\ \mu C, \frac{800}{3}\ \mu C$

(D) $\frac{1000}{3}\ \mu C, \frac{1400}{3}\ \mu C$

Space for Rough Work

Paragraph for Questions 13 and 14

A light body revolving around a heavier body due to gravitational attraction, is called satellite. Earth is a satellite of the Sun while Moon is satellite of Earth.

Orbital velocity (v_0) : A satellite of mass m moving in an orbit of radius r with speed v_0 then required centripetal force is provided by gravitation.

$$F_{cp} = F_g \Rightarrow \frac{mv_0^2}{r} = \frac{GMm}{r^2} \Rightarrow v_0 = \sqrt{\frac{GM}{r}} = \sqrt{\frac{GM}{(R_e + h)}}, (r = R_e + h)$$

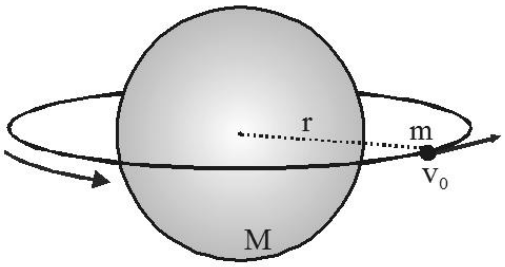
For a satellite very close to the Earth surface $h << R_e \therefore r = R_e$

$$v_0 = \sqrt{\frac{GM}{R_e}} = \sqrt{gR_e} = 8 \text{ km/s}$$

Essential Condition's for Satellite Motion

Centre of satellite's orbit coincide with centre of Earth.

Plane of orbit of satellite is passing through centre of Earth.



3. A near earth orbit satellite of earth experiences very small air resistance. It's period of revolution decreases by 1.08 sec/day.

(A) It's radius decreases by $\frac{160}{3}$ m in 1 day.

(B) It's radius decreases by $\frac{160}{3}$ m in 1 revolution.

(C) It's angular momentum decreases

(D) It's mechanical energy increases.

Space for Rough Work

4. Satellite A is in a circular orbit of radius r and satellite B in another circular orbit of radius 4 r, both revolving around the earth. The masses of the satellites A and B are in the ratio 3 : 1. If the symbols V, E, u and T represent the speed, total energy, escape velocity and period of revolution of satellite, then

(A) $V_A/V_B = 2$ (B) $E_A/E_B = 12$ (C) $u_A/u_B = 2$ (D) $T_A/T_B = \frac{1}{8}$

Space for Rough Work

Paragraph for Questions 15 and 16

Electric potential is a scalar property of every point in the region of electric field. At a point in electric field potential is defined as the interaction energy of a unit positive charge. If at a point in electric field a charge q_0 has potential energy U, then electric potential at that point can be given as

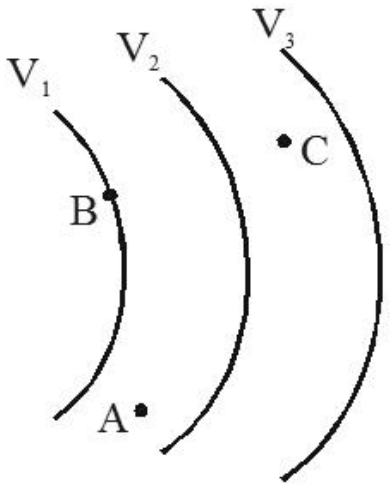
$V = \frac{U}{q_0}$ joule/coulomb

Potential difference between two points in electric field can be defined as work done in displacing a unit positive charge from one point to another against the electric forces.

If a unit +ve charge is displaced from a point A to B as shown work required can be given as

$V_B - VA = -\int_A^B \vec{E}.\vec{dr}$

5. Consider three equipotential surfaces of field produced by –q point charge. A small point charge +q is taken slowly along two paths from A to B and A to C, then :-



(A) Work done by electric field in moving it from A to B is +ve

(B) Work done by electric field in moving it from A to B is –ve

(C) Work done by an external agent in moving it from A to C is +ve

(D) The correct order of potential will be $V_3 > V_2 > V_1$.

Space for Rough Work

6. The potential of electric field $V = x^2y$, where x and y denotes coordinate of a point in electric field region. Which of the following is incorrect.

(A) Electric field at all points on positive x-axis is in positive y-direction

(B) Electric field at all points on positive y-axis is in negative x-direction.

(C) Electric field at origin is non-zero.

(D) Electric field at (x, y) makes an angle of $\theta = \cot^{-1}\left(\frac{2y}{x}\right)$ with x-axis.

Space for Rough Work

SEC-3 : (Maximum Marks: 16)

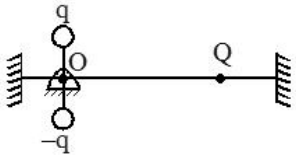
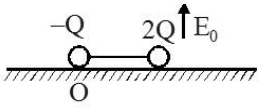
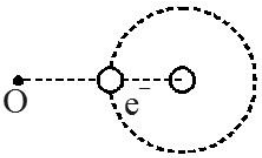
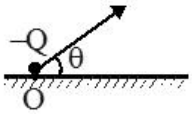
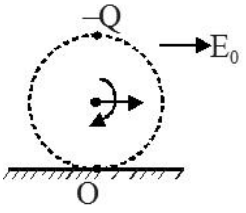
This section contains **TWO (02)** questions.
Each question contains two columns, **Column-I** and **Column-II**.
Column-I has **four** entries (A), (B), (C) and (D)
Column-II has **five** entries (P), (Q), (R), (S) and (T)
Match the entries in **Column-I** with the entries in **Column-II**.
One or more entries in **Column-I** may match with one or more entries in **Column-II**.
For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then chosen these three options. Similarly, for entries (B), (C) and (D).
For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks	:	+2	If only the option(s) corresponding to all the correct match(es) is (are) chosen
Zero Marks	:	0	If none of the options is chosen
Negative Marks	:	–1	In all other cases

Space for Rough Work

1.

Column-I		Column-II	
(A)	Magnitude of angular momentum of system about O increases after the instant shown.	(P)	A dipole is hinged on a long smooth fixed horizontal wire. A charge Q (A) is threaded on wire and is free to move on the wire. The system comprises of the dipole and point charge. Gravity is absent. 
(B)	The net torque on body A is zero about O at the instant shown.	(Q)	A system of two charges –Q and 2Q (A) of mass m each are connected by a light rod and kept on smooth level ground.An external uniform electric field E_0 = mg/Q is applied as shown. Gravity is present. 
(C)	The net force on negative charge increases in magnitude after the instant shown.	(R)	An electron (A) is revolving about a fixed proton in anticlockwise direction The system comprises of the electron and proton. Gravity is absent. 
(D)	Mechanical energy of the system increases after the instant shown. (ignore potential energy of external electrical field.)	(S)	System consists of a particle (A) of mass m and charge –Q. It is projected from level ground in presence of constant electric field E = mg/Q in vertically upward direction. Gravity is present. 
		(T)	System consists of a uniform ring and particle (A) of mass m and charge –Q on its circumference as shown. The ring is in pure rolling on a sufficiently rough horizontal surface. Gravity is present. 

Space for Rough Work

2. In column-I, we have certain situations where a battery of variable voltage V is connected in a circuit. Column-II gives comments on current in element A on variation of voltage :-

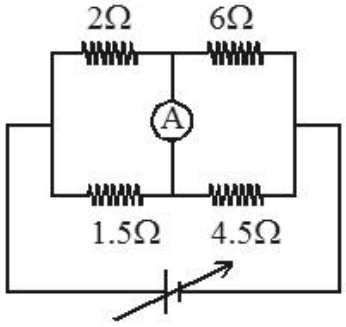
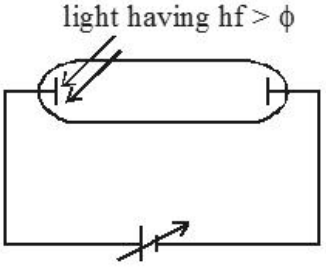
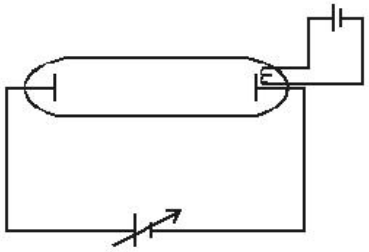
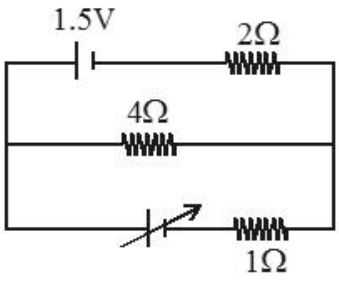
<table>
<tr><th colspan="2">Column-I</th><th colspan="2">Column-II</th></tr>
<tr><td>(A)</td><td> 
</td><td>(P)</td><td>on increasing V, i may/will increase</td></tr>
<tr><td>(B)</td><td> 

photo electric setup
A → photoelectric tube</td><td>(Q)</td><td>On increasing V, i may/will decrease</td></tr>
<tr><td>(C)</td><td> 
A → x ray tube</td><td>(R)</td><td>On increasing V, i may/will not change at all</td></tr>
<tr><td>(D)</td><td> 

A → 4Ω resistor</td><td>(S)</td><td>On reversing V, i may/will be zero</td></tr>
<tr><td colspan="2"></td><td>(T)</td><td>On reversing V, i may/will increase</td></tr>
</table>

Space for Rough Work

CHEMISTRY

SEC-1 : (Maximum Marks: 40)

- This section contains **TEN (10)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

Space for Rough Work

1. 10 gm of an impure sample of $CaCO_3$ on heating gives 4.48 gm of solid residue. Assuming that all $CaCO_3$ is completely decomposed into CaO and CO_2, identify correct statement(s) {Ca = 40}

(A) The impurity is non-volatile

(B) Impurity is volatile

(C) Percent purity of $CaCO_3$ sample is 80%

(D) CO_2 liberated is 3.52 gm

2. The following four reactions are at equilibrium in closed rigid containers. In which of the reaction(s) equilibrium is not disturbed by adding He gas.

(A) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

(B) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

(C) $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$

(D) $C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$

3. A and B are two volatile liquids with their vapour pressure : $P_A^0 = 340mm$ of Hg and $P_B^0 = 420mm$ of Hg at 25°C. To a flask containing 8 mole of 'A' , 5 mole of 'B' was added. As soon as 'B' was added , 'A' starts associating to form a non-volatile solid A_x(solid) which is soluble in both liquids A and B. The association of 'A' follows 1st order kinetics. The vapour pressure of solution measured after 1 hour is 360 mm of Hg and after very long time it is 300 mm of Hg. Identify correct statements(s)

(A) The value of x = 4

(B) At t = 1 hr, 0.5 mole of A_x (solid) is formed

(C) At t = 1 hr, 4 mole of 'A' (liquid) remains in solution.

(D) The value of x = 2

4. Which of the following order(s) is/are correct?

(A) Al > Ga (Atomic Size)

(B) $P^{-3} > Ca^{+2}$ (Ionic Size)

(C) Ne > Al (Electron Affinity)

(D) Li > B (First Ionisation Energy)

Space for Rough Work

5. Select the reaction(s) which is/are not possible.

(A) $2NaF + 2H_2O \rightarrow 2NaOH + H_2 + F_2$

(B) $\underset{\text{fused}}{KHF_2} \xrightarrow{\text{electrolysis}} H_2 + F_2$

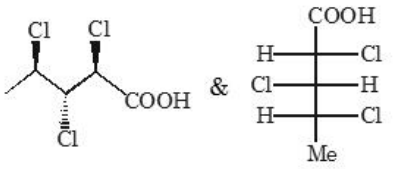
(C) $NaCl + H_2O \rightarrow NaOH + HCl$

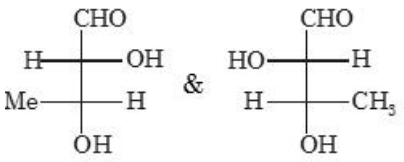
(D) $\underset{\text{(hot and conc.)}}{2NaOH} + Cl_2 \longrightarrow NaCl + NaOCl + H_2O$

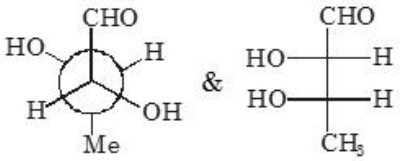
6. Which of the following element(s) undergoes disproportionation in basic medium?

(A) P_4 (B) Cl_2 (C) S_8 (D) Si

7. Which of the following pair of compounds doesn't represent enantiomeric pair :

(A) 

(B) 

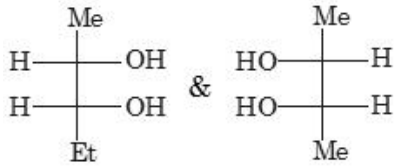
(C) 

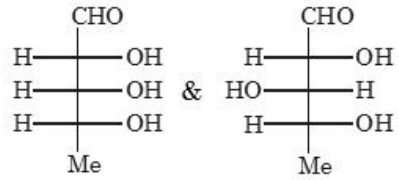
(D) 

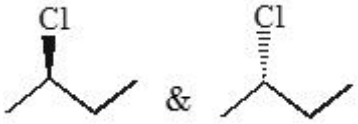
Space for Rough Work

8. Identify pair(s) of compounds which can be seperate by fractional distillation ?

(A) 

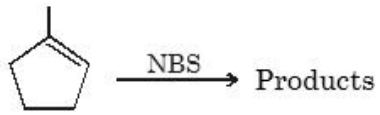
(B) 

(C) 

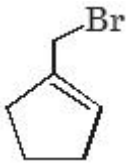
(D) 

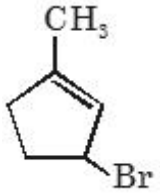
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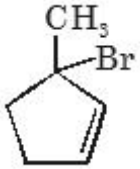
9.

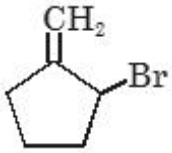


Possible product(s) is/are :

(A) 

(B) 

(C) 

(D) 

10. Correct statement :

(A) Vinyl chloride > Benzyl chloride (C–Cl bond length)

(B) Allyl bromide > Vinyl bromide (Reactivity towards S_N2)

(C) Tertiary butyl bromide > Secondary butyl bromide (Reactivity towards S_N1)

(D) 7-Chloro 1,3,5-cycloheptatriene > 5-chloro 1,3-cyclopentadiene (Rate of hydrolysis)

Space for Rough Work

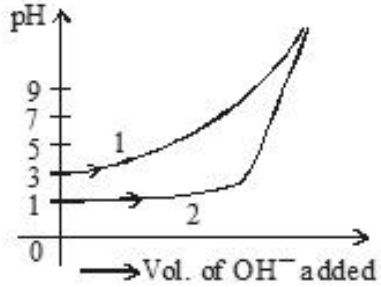
SEC-2 : (Maximum Marks: 24)

This section contains **SIX (06)** questions.
This section contains **THREE** paragraphs.
Based on each paragraph, there are **TWO** questions
Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
For each question, choose the option(s) corresponding to (all) the correct answer(s)
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –2 In all other cases.
For Example : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

Space for Rough Work

Paragraph for Questions 11 and 12

Titration of acid samples with hydroxide ion solution produces very different pH curves depending upon the type and strength of acid. Given below are two titration curves for two different monobasic acid having same initial concentration of 0.1M



1. With respect to above information in paragraph indentify correct statement(s)

(A) Curve (1) may be of HCl while (2) may be of CH_3COOH.

(B) The nature of solution at equivalence point for curve(1) is acidic

(C) Phenolphthalein can be used to detect the equivalence point for curve (1)

(D) For weaker acid among (1) and (2) the value of $K_a = 10^{-5}$

2. Let the colour of an acidic indicator HIn (colourless) be visible only when it's ionized form (pink) is 25% or more than the unionized form in a solution. If HIn ($pK_a = 9$) is added to solution of pH = 9.6, then correct observation(s) is/are -

(A) Pink colour will be visible

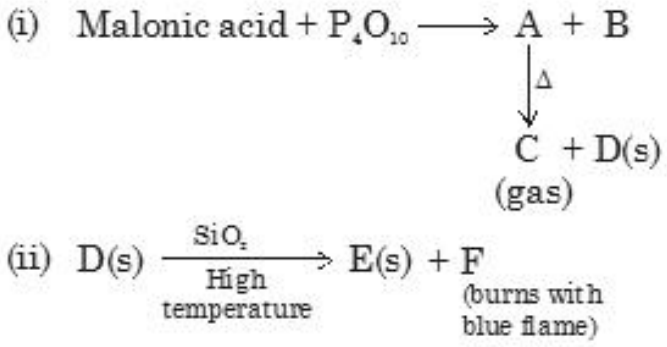
(B) Pink colour will not be visible

(C) Unionized form of indicator is less than 25% in solution

(D) Ionized form of indicator is less than 25% in solution

Space for Rough Work

Paragraph for Questions 13 and 14



3. The gas C can be differentiate from SO_2 by using of :-

(A) $KMnO_4/H^+$ (B) $K_2Cr_2O_7/H^+$ (C) KIO_3 + starch (D) KOH(aq.)

4. Which of the following is/are correct for E(s)?

(A) It reacts with D to form a very hard substance used as an abrasive

(B) Most abundant element in earth crust

(C) It reacts with oxygen at 1173 K to form a 3D silicate

(D) It can be purified by zone refining

Space for Rough Work

Paragraph for Questions 15 and 16

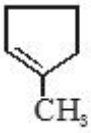
Compound A_1, A_2, A_3 having molecular formula C_6H_{10}. All have five membered ring. A_1, A_2, A_3 are strutural isomers.

$A_1 \xrightarrow[(CCl_4)]{Br_2} \rightarrow$ Enantiomeric pair is formed as product

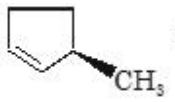
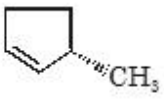
$A_2 \xrightarrow[(CCl_4)]{Br_2} \rightarrow$ Diastereomeric pair is formed as product

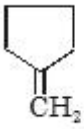
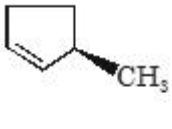
$A_3 \xrightarrow[(CCl_4)]{Br_2} \rightarrow$ Non-resolvable optically inactive product is formed

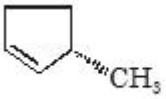
5. A_1 can be :

(A)  (B)  (C)  (D) All

6. A_2 & A_3 may respectively :

(A)  ; 

(B)  ; 

(C)  ; 

(D)  ; 

Space for Rough Work

SEC-3 : (Maximum Marks: 16)

This section contains **TWO (02)** questions.
Each question contains two columns, **Column-I** and **Column-II**.
Column-I has **four** entries (A), (B), (C) and (D)
Column-II has **five** entries (P), (Q), (R), (S) and (T)
Match the entries in **Column-I** with the entries in **Column-II**.
One or more entries in **Column-I** may match with one or more entries in **Column-II**.
For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then chosen these three options. Similarly, for entries (B), (C) and (D).
For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the option(s) corresponding to all the correct match(es) is (are) chosen
Zero Marks : 0 If none of the options is chosen
Negative Marks : –1 In all other cases

---

1. Match the column :

	Column I		Column II
(A)	$CH_3COCH_3 + CHCl_3$	(P)	$\Delta S_{mix} > 0$
(B)	$C_2H_5OH + H_2O$	(Q)	$\Delta V_{mix} > 0$
(C)	$C_2H_5Br + C_2H_5I$	(R)	$\Delta H_{mix} < 0$
(D)	$CH_3COCH_3 + CS_2$	(S)	Maximum boiling azeotrope
		(T)	Minimum boiling azeotrope

Space for Rough Work

2. Match the Column.

	Column-I Compounds		**Column-II Characteristics**
(A)	Tetrachloroethene	(P)	Liquefaction of it is due to London forces only
(B)	NO_2	(Q)	Used as a solvent for dry cleaning
(C)	NO	(R)	At least one atom is sp^2 hybridized
(D)	CO	(S)	Bond order of at least one bond is greater than 2
		(T)	Odd electron species

Space for Rough Work

MATHEMATICS

SEC-1 : (Maximum Marks: 40)

- This section contains **TEN (10)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

1. If the pair of lines $x^2 + 2xy + ay^2 = 0$ and $ax^2 + 2xy + y^2 = 0$ have exactly one line in common and the joint equation of the other two lines is $\alpha x^2 + \beta xy + \gamma y^2 = 0$ where $\alpha,\beta,\gamma \in N$, then lowest value of

(A) $\alpha = 3$ (B) $\beta = 9$ (C) $\alpha + \gamma = 6$ (D) $\alpha + \beta = 13$

Space for Rough Work

2. Two vertices Q and R of a ΔPQR are mirror images of each other with respect to the line y = x. Q is (5,4) and the orthocentre of ΔPQR is origin. If P is (a,b) then [a + b] is greater than
[Note : [ ] denotes greatest integer function]

(A) 6 (B) 7 (C) 8 (D) 9

3. The equation of a plane which is equally inclined to the lines $\frac{x-1}{2} = \frac{y}{-2} = \frac{z+2}{-1}$ and $\frac{x+3}{8} = \frac{y-4}{1} = \frac{z}{-4}$ and passing origin is/are

(A) $14x - 5y - 7z = 0$ (B) $3x - 4y - z = 0$

(C) $x + 2y - 5z = 0$ (D) $2x + 7y - z = 0$

4. Let the intercept made by the line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ between the plane x + y + z = 3 and the line $\frac{x-1}{2} = \frac{y-2}{-1} = \frac{z-3}{-1}$ is k, then

(A) $\log_{14}(2k) = \frac{1}{3}$ (B) $\log_{14}(2k) = \frac{1}{2}$

(C) $\log_2(2k) \in (2,3)$ (D) $\log_2(2k) \in (3,4)$

5. Let $\vec{p} = x^2\hat{i} - 3\hat{j} + (x+3)\hat{k}$ and $\vec{q} = \hat{i} + 3\hat{j} - (x-3)\hat{k}$ be two vectors such that $|\vec{p}| = |\vec{q}|$. If $\vec{r} = 2\vec{p} + 3\vec{q}$ and $\vec{s} = 3\vec{p} - 2\vec{q}$, then

(A) $\int_1^3 (\vec{r} \wedge \vec{s})\,dx = \frac{\pi}{2}$ (B) $\int_1^3 (\vec{r} \wedge \vec{s})\,dx = \pi$

(C) $\int_2^{5/2} (\vec{r} \wedge \vec{s})\,dx = \frac{\pi}{2}$ (D) $\int_2^{5/2} (\vec{r} \wedge \vec{s})\,dx = \frac{\pi}{4}$

Space for Rough Work

6. If the curves $x^2 - y^2 = 4$ and $xy = \sqrt{5}$ intersects at points A and B, then the possible number of point(s) C on curve $x^2 - y^2 = 4$ such that ΔABC is equilateral is less than or equal to-

(A) 0 (B) 1 (C) 2 (D) 4

7. If number of integral points (α,β) (an integral point is a point whose both co-ordinates are integers) for which the area common to the circle $x^2 + y^2 + 2\alpha^2 x + 2\beta^2 y + c = 0$ and its image in the line $x + y + 4 = 0$ is maximum is k, then k is greater than or equal to-

(A) 3 (B) 4 (C) 5 (D) 6

8. 2 parabolas have the focus (5,–2). Their directrices are x-axis and y-axis respectively. If the slope of their common chord is m then m^2 is less

(A) 5 (B) 3 (C) $\sqrt{2}$ (D) 1

9. If the planes $\vec{r}.(\hat{i} + \hat{j} + \hat{k}) = q_1, \vec{r}.(\hat{i} + 2a\hat{j} + \hat{k}) = q_2$ and $\vec{r}.(a\hat{i} + a^2\hat{j} + \hat{k}) = q_3$ intersect in a line then the value of 'a' is(are)

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) 1

10. Let $\vec{a}$ and $\vec{b}$ are two vectors such that $\vec{a} = 2(\sin\alpha + \cos\alpha)\hat{i} + \hat{j}$ and $\vec{b} = (\sin\beta)\hat{i} + (\cos\beta)\hat{j}$ where α and β satisfy the relation $2(\sin\alpha + \cos\alpha)\sin\beta = 3 - \cos\beta$, then (where $\alpha, \beta \in R$ and $\alpha \in [0, \frac{\pi}{2})$)

(A) $2\tan^2\alpha + \tan^2\beta = 9$ (B) $\tan^2\alpha + 2\tan^2\beta = 9$

(C) $\tan^2\alpha + 4\sqrt{2}\tan\beta = 17$ (D) $\tan^2\alpha - \tan^2\beta = 1$

Space for Rough Work

SEC-2 : (Maximum Marks: 24)

This section contains **SIX (06)** questions.
This section contains **THREE** paragraphs.
Based on each paragraph, there are **TWO** questions
Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
For each question, choose the option(s) corresponding to (all) the correct answer(s)
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –2 In all other cases.
For Example : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

Space for Rough Work

Paragraph for Questions 11 and 12

The circle $C_1 : x^2 + y^2 = r^2$ and ellipse $C_2 : \frac{x^2}{16} + \frac{y^2}{9} = 1$ intersect at 4 distinct points A,B,C,D. Their common tangents form a parallelogram A'B'C'D'.

1. If A'B'C'D' is a square then r is equal to $\frac{a}{\sqrt{b}}$ where a and b are coprime then -

(A) $a + b = 12$ (B) $a + b = 7$ (C) $a + 2b = 9$ (D) $a + b^2 = 9$

2. If A'B'C'D' is a square, then the ratio of area of the circle C_1 to the area of the circumcircle of $\Delta A'B'C'$ is K, then K is greater than or equal to

(A) $\frac{9}{16}$ (B) $\frac{3}{4}$ (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

Paragraph for Questions 13 and 14

Let A,B,C be 3 sets such that

$A = \{(x, y) : \frac{x}{\cos\theta} = \frac{y}{\sin\theta} = 2, \theta \in R\}$

$B = \{(x, y) : \frac{x+1}{\cos 2205^\circ} = \frac{y}{\sin 2205^\circ} = r, r \in R\}$

$C = \{(x, y) : (x - \sqrt{3})^2 + (y - 1)^2 \leq a^2; a \in R\}$

Let C_1,C_2 be the curves represented by A and B respectively while R be the region represented by C.

3. A points $P(x_1,y_1)$ and $Q(x_2,y_2)$ be such that $P,Q \in A \cap B$, then point of intersection of tangents drawn to C_1 at P and Q is-

(A) (2,3) (B) (–4,4) (C) (3,–3) (D) (–5,5)

Space for Rough Work

4. If minimum value of 'a' for which $A \cap C = A$ is α, then length of curve C_2 lying in R corresponding to $a = \alpha$, is

(A) $\sqrt{50}$ (B) $\sqrt{52}$ (C) $\sqrt{58}$ (D) $\sqrt{57}$

Paragraph for Questions 15 and 16

Let $P_1 : x + y + 2z - 3 = 0$ and $P_2 : x - 2y + z = 4$ be two planes. Let A(1,3,4) and B(3,2,7) be two points in R^3.

5. The equation of plane P_3 through line of intersection of P_1 and P_2 and length of projection upon which of the line segment AB is greatest is equal to $ax + by + cz + 1 = 0$, then
(where $a,b,c \in I$)

(A) $a + b + c = 4$ (B) $a + 2b + c = 7$

(C) $a - b + c = 0$ (D) $a + 2b + c = 5$

6. The equation of plane P_4 through the line of intersection of P_1 and P_2 and the length of projection upon which of the line segment AB is least is

(A) $2x - y + 3z - 7 = 0$ (B) $3y + z + 1 = 0$

(C) $x + 3y + z + 2 = 0$ (D) $3x - 3y + 4z - 11 = 0$

Space for Rough Work

SEC-3 : (Maximum Marks: 16)

This section contains **TWO (02)** questions.
Each question contains two columns, **Column-I** and **Column-II**.
Column-I has **four** entries (A), (B), (C) and (D)
Column-II has **five** entries (P), (Q), (R), (S) and (T)
Match the entries in **Column-I** with the entries in **Column-II**.
One or more entries in **Column-I** may match with one or more entries in **Column-II**.
For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then chosen these three options. Similarly, for entries (B), (C) and (D).
For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the option(s) corresponding to all the correct match(es) is (are) chosen
Zero Marks : 0 If none of the options is chosen
Negative Marks : –1 In all other cases

---

1. Let two lines be $L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-\alpha}{4}$ and $L_2 : \frac{x-3}{\beta} = \frac{y-5}{6} = \frac{z-5}{8}$, then

Column-I		Column-II	
(A)	If $\beta = 4$ and $\alpha \in R - \{1\}$	(P)	Non coplanar lines
(B)	If $\beta = 4$ and $\alpha = 1$	(Q)	Coplanar lines
(C)	If $\beta \in R - \{4\}$ and $\alpha = 1$	(R)	Lines lie in a unique plane
(D)	If $\beta \in R - \{4\}$ and $\alpha \in R - \{1\}$	(S)	Infinite planes containing both the lines
		(T)	Lines do not intersect

Space for Rough Work

2.

Column-I		Column-II	
(A)	If a variable line $y = ax + b$ is tangent to ellipse $3x^2 + 4y^2 = 6$, then locus of point P(a,b) is a conic whose eccentricity is equal to	(P)	$\frac{\sqrt{3}}{2}$
(B)	Consider three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ such that $\lvert\vec{a}\rvert = \sqrt{3}, \vec{a}.\vec{c} = 2$ and $\vec{a} \times \vec{c} = \vec{b}$, then value of $\frac{1}{8}\lvert\vec{a} \times \vec{b} + 3\vec{c} + 2\vec{a}\rvert$ is equal to	(Q)	$\sqrt{\frac{3}{2}}$
(C)	Let A and B denote the points of intersection of the circles $x^2 + y^2 - 6x + 4y = 3$ and $x^2 + y^2 + 4x - 4y = 17$, then slope of line AB is equal to	(R)	1
(D)	Locus of the point of intersection of the pair of perpendicular tangents to the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 7$ is a circle of radius 'r' then 'r' is greater than	(S)	2
		(T)	$\frac{5}{4}$

Space for Rough Work

Space for Rough Work

PHYSICS

SEC-A : (Maximum Marks: 80)

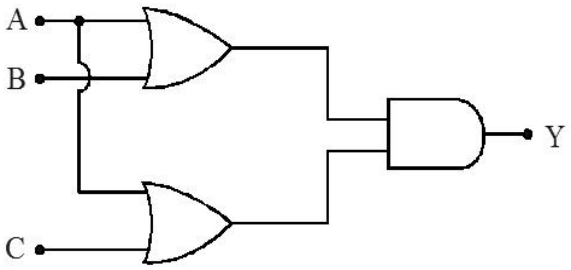
This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : –1 If wrong option is selected.

1. The output of given logic circuit is :-



(A) $A \cdot (B + C)$ (B) $A \cdot (B \cdot C)$

(C) $(A + B) \cdot (A + C)$ (D) $A + B + C$

2. Which one of the following statements is wrong?

(A) Radio waves in the frequency range 10MHz to 20 MHz are called sky waves

(B) Radio horizon of the transmitting antenna for space wave is $d_T = \sqrt{(Rh_T)}$ (R = radius of earth, h_T = height of transmitting antenna)

(C) Fiber optical communication is almost free from electrical disturbances

(D) The principle of fibre optical communication is total internal reflection

3. Three copper rods are subjected to different potential difference. Compare the drift speed of electrons through them. Assume that all 3 are at the same temperature.

	Length	Diameter	Potential difference
(A)	L	3d	V
(B)	2L	d	2V
(C)	3L	2d	2V

(A) $v_A = v_B > v_C$ (B) $v_A > v_B > v_C$

(C) $v_A < v_B < v_C$ (D) None of these

4. Three infinitely long linear charges of charge density λ, λ and -2λ are placed in space. A point in space is specified by its perpendicular distance r_1, r_2 & r_3 respectively from the linear charges. For the points which are equipotential :-

(A) $\frac{r_1 r_2}{r_3^2}$ = constant

(B) $r_1 r_2 r_3^2$ = constant

(C) $r_1 r_2 r_3^{1/2}$ = constant

(D) $r_1 r_2 r_3$ = constant

5. Choose the correct option :-

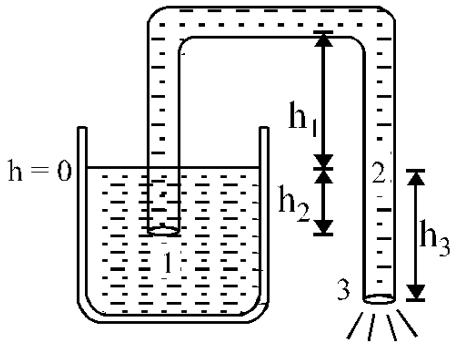
(A) Work done by external agent to take negative charge from higher potential to lower potential is negative.

(B) Work done by external agent to take positive charge from lower potential to higher potential is positive.

(C) Work done by electric field on negative charge for motion from higher potential energy to lower potential energy is negative.

(D) Work done by electric field on positive charge for motion from lower potential energy to higher potential energy is positive.

6. Figure shows a siphon. Choose the wrong statement: (P_0 = atmospheric pressure)



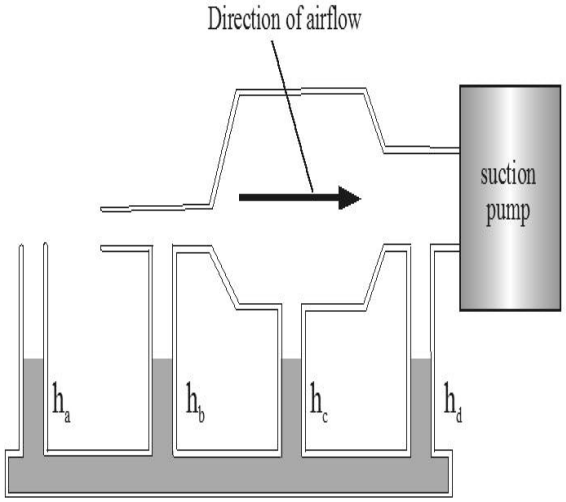
(A) Siphon works when $h_3 > 0$

(B) Pressure at point 2 is $P_2 = P_0 - \rho g h_3$

(C) Pressure at point 3 is P_0

(D) Pressure at point 1 is P_0

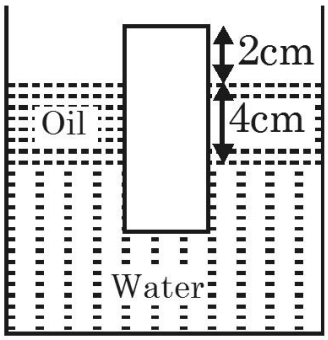
7. Rank in order, from highest to lowest, the liquid heights h_a to h_d. The air flow is from left to right. The liquid columns are not drawn to scale :-



(A) $h_a > h_b = h_d > h_c$ (B) $h_b = h_d > h_c > h_a$

(C) $h_d > h_c > h_b > h_a$ (D) $h_b > h_d > h_c > h_a$

8. A cylinder of radius 4cm and height 10cm is immersed in two liquids as shown. Specific gravity of oil is 0.5. 2cm of cylinder is in the air. Select the incorrect statement. Neglect atmospheric pressure.



(A) Force exerted by oil on the cylinder is zero.

(B) Force exerted by water on the cylinder is $0.96\ \pi$ newton.

(C) Force exerted by both water and oil is $1.14\ \pi$ newton.

(D) Frequency of oscillation does not depend on density of oil, if cylinder is displaced slightly.

9. A 2 μF capacitor is charged to a potential = 10V. Another 4 μF capacitor is charged to a potential = 20V. The two capacitors are then connected in a single loop, with the positive plate of one connected with negative plate of the other. What heat is evolved in the circuit ?

(A) 300 μJ (B) 600 μJ

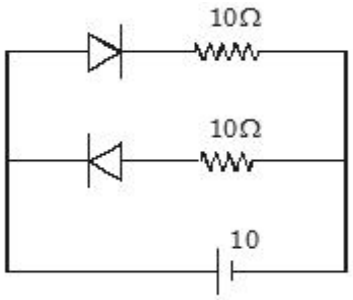
(C) 900 μJ (D) 450 μJ

10. Time period of simple pendulum increases by an amount $\sqrt{2}$ times at height 'h' from the surface of earth. Then the value of h is :

(A) 0.25 R (B) 0.4 R

(C) 0.5 R (D) R

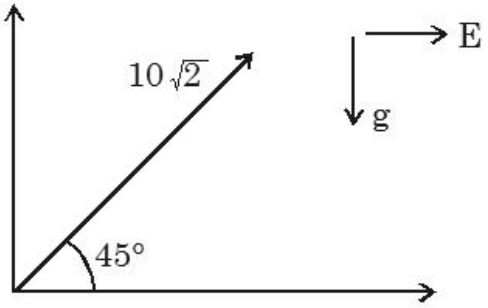
11. Reverse break down voltage for silicon diode is 6V for a particular doping concentration and forward break down voltage is 0.3V then find approximate current through battery in the given circuit



(A) 2 Amp (B) 1.3 Amp

(C) 1 Amp (D) $\frac{\pi}{\sqrt{2}}$ Amp

12. A particle of mass 1gm and charge –0.1μC is projected from ground with a velocity $10\sqrt{2}$ at an 45° with horizontal in the area having uniform electric field 1 kV/cm in horizontal direction. Acceleration due to gravity is $10m/s^2$ in vertical downward direction. Select incorrect statement:



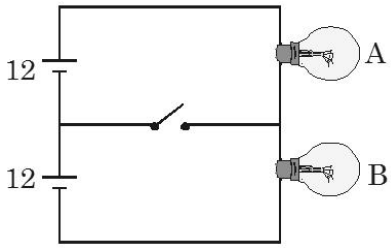
(A) Time of flight for particle is 2 sec

(B) Range of particle is 20 m

(C) Total displacement of particle is 0m

(D) Particle will follow straight line motion.

13. Two light bulbs in the circuit have rating A [24 V, 48W] and B [24V, 36W] as shown.



(A) When switch is closed bulb A delivers 24 watt & B delivers 18 watt.

(B) When switch is open A delivers 9 watt & B delivers 12 watt approximately.

(C) When switch is open A delivers 24 watt & B delivers 18 watt approximately.

(D) When switch is open 'A' delivers 6 watt and B delivers 9 watt approximately.

14. A 0.5 kg mass of lead is submerged in a container filled to the brim with water and a block of wood floats on top. The lead mass is slowly lifted from the container by a thin wire and as it emerges into air the level of the water in the container drops a bit. The lead mass is now placed on the block of wood. As the lead is placed on the wood.

(A) Some water spills over the edge of the container.

(B) The water level rises exactly to the brim as before.

(C) The water level rises but does not reach the brim of the container

(D) There is not enough information provided to decide between the three options.

15. A spherical asteroid having the same density as that of earth is floating in free space. A small pebble is revolving around the asteroid under the influence of gravity near the surface of the asteroid. What is the approximate time period of the pebble?

(A) 24 h (B) 365 days

(C) 10 min (D) 1 hr 24 min

16. A transistor is connected in common emitter (CE) configuration. The collector supply is 8 V and the voltage drop across a resistor of 800 Ω in the collector circuit is 0.5 V. If the current gain factor α is 0.96, find the base current.

(A) 48 μA (B) 44 μA

(C) 26 μA (D) 15 μA

17. Carbon resistor has resistance specified by three bands having colour red, yellow and black. If this resistor is cut into two pieces of equal length then the new colour code of each one will be (Neglect tolerance of 4th band)

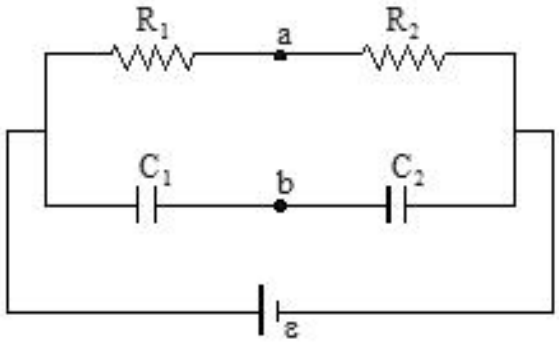
(A) Brown, Red, Black

(B) Red, Orange, Black

(C) White, Blue, Black

(D) Black, Blue, Black

18. Two resistors with resistances R_1 and R_2 are connected in series and so are two capacitors with capacitances C_1 and C_2, these two systems are connected to a battery as shown in figure. Potential difference between a and b is zero in steady state if



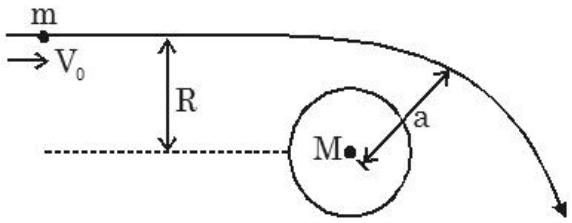
(A) $R_1 + R_2 = \dfrac{C_1C_2}{C_1 + C_2}$

(B) $R_1R_2 = C_1C_2$

(C) $R_1C_2 = R_2C_1$

(D) $R_1C_1 = R_2C_2$

19. A mass m, travelling at speed V_0 in a straight line from far away is deflected when it passes near a black hole of mass M which is at a perpendicular distance R from the original line of flight. a, the distance of closest approach between the mass and the black hole is given by the relation :



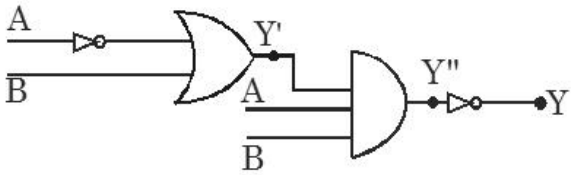
(A) $a = R(1 + \frac{2GM}{aV_0^2})^{1/2}$

(B) $a = R(1 + \frac{aV_0^2}{2GM})^{1/2}$

(C) $a = R(1 + \frac{GM}{2aV_0^2})^{-1/2}$

(D) $a = R(1 + \frac{2GM}{aV_0^2})^{-1/2}$

20. Circuit given below is that of



(A) AND gate

(B) OR gate

(C) NAND gate

(D) NOR gate

SEC-B : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

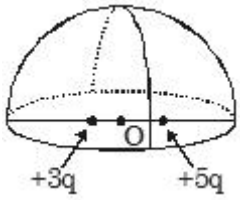
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

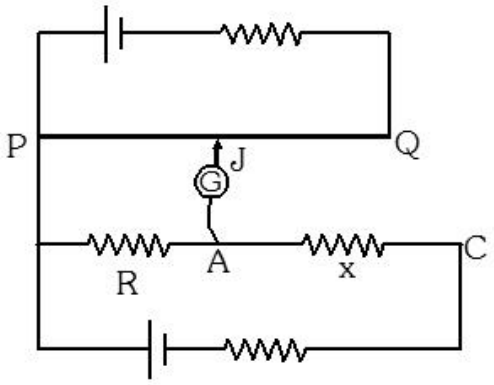
Zero Marks : 0 In all other cases.

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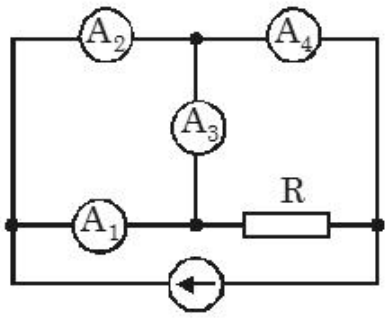
1. The figure shows a hollow hemisphere of radius R in which two charges 3q and 5q are placed symmetrically about the centre O on the planar surface. The electric flux over the curved surface is $\frac{nq}{\varepsilon_0}$. Fill the value of n.



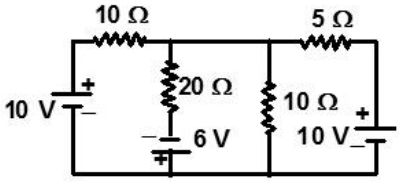
2. Circuit for the measurement of resistance by potentiometer is shown. The galvanometer is first connected at point A and zero deflection is observed at length P J = 10 cm. In second case it is connected at point C and zero deflection is observed at a length 30cm from P. Then the unknown resistance X is nR, fill the value of n.



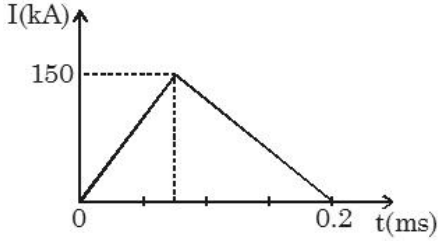
3. Four ammeters with identical internal resistances r and a resistor of resistance R are connected to a current source as shown in figure. It is known that the reading of the ammeter A_1 is I_1 = 3 A and the reading of the ammeter A_2 is I_2 = 5 A. Determine the ratio of the resistances R/r.



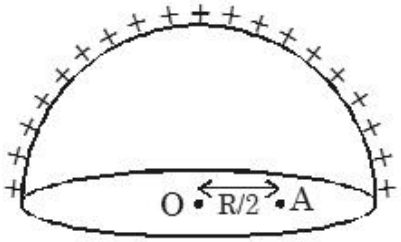
4. The value of current through the 20Ω resistor is $\frac{n}{10}$ amp. Fill the value of n.



5. During lighting, a current pulse, shown in figure, flows from the cloud at a height 1.5km to the ground. If the breakdown electric field of humid air is about 400 kVm^{-1}, the energy released during lighting would be (in unit of 10^9 J)



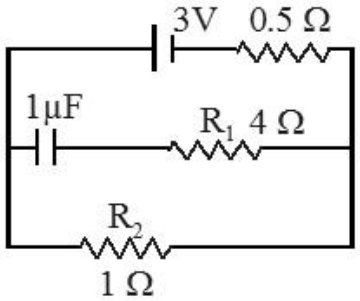
6. A hemispherical bowl of mass m is uniformly charged with charge density 'σ'. Electric potential due to charge distribution at a point 'A' is $\frac{\sigma R}{n\varepsilon_0}$, fill the value of n. (which lies at centre of radius as shown).



7. Dipole having dipole moment 4C-m is placed on origin along x-axis. Point charge of 8μC is fixed at (4, 0, 0) m. Now dipole is rotated by an angle of $\frac{\pi}{2}$. If the work done (in kJ) in rotating the dipole is 6x, then fill the value of x.

8. A soap bubble in a form of circular tube having radius of curvature R and radius of curvature perpendicular to it is 5R. If the excess pressure in the bubble is $\frac{3nT}{5R}$, then fill the value of n.

9. A 1μF capacitor is connected in the circuit shown below. The emf of the cell is 3 volts and internal resistance is 0.5 ohms. The resistors R_1 and R_2 have values 4 ohms and 1 ohm respectively. The charge (in μC) on the capacitor in steady state must be :



10. A parallel plate capacitor of capacitance C is connected to a battery and is charged to a potential difference V. Another capacitor of capacitance 2C is similarly charged to a potential difference 2V. The charging battery is now disconnected and the capacitors are connect in parallel to each other in such a way that the positive terminal of one is connected to the negative terminal of the other. The final energy of the configuration is $\frac{nCV^2}{2}$, then fill the value of n.

SEC-A : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : –1 If wrong option is selected.

1. Observe the following four solutions :

P : 10 ml, 0.1 M NaOH + 5ml, 0.1 M HCl

Q : 10 ml, 0.1 M NaOH + 15ml, 0.1 M CH_3COOH

R : 10 ml, 0.1 M NH_3 + 10ml, 0.1 M NH_4Cl

S : 10 ml, 0.05 M NaF + 5ml, 0.1 M HF

Which of above solutions act as buffer

(A) P and Q only (B) R and S only

(C) P, R, S only (D) Q, R, S only

2. Which of the following air pollutant is likely not to produce photo chemical smog?

(A) NO (B) CH_2CHCHO

(C) $CH_3\overset{\overset{O}{\|}}{C}OONO_2$ (D) CO_2

3. Which of the following is false regarding acid rain.

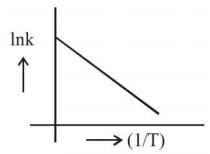
(A) Acid rain has pH below 5.6

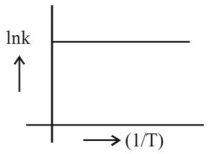
(B) CO_2 and CO are major contributors of acid rain

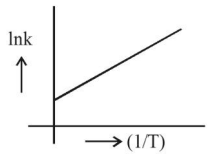
(C) Acid rain washes away plant nutrients needed for their growth

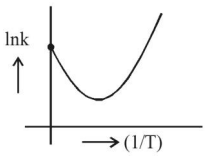
(D) Acid rain is responsible for the change of colour of Tajmahal.

4. An equilibrium shift towards reactants at higher temperatures. Find the correct graph :

(A) 

(B) 

(C) 

(D) 

5. Which of the following properties is/are true for hydrogen peroxide?

(i) Mild disinfectant

(ii) Rocket propellant

(iii) Bleaching agent

(iv) Oxidising agent

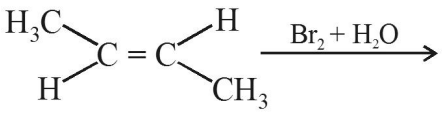
(A) only (iii), (iv)

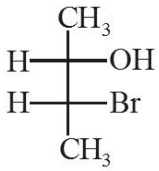
(B) only (i), (iii) (iv)

(C) only (i), (iv)

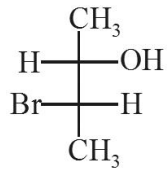
(D) All (i), (ii), (iii) & (iv)

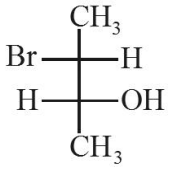
6. Major product of reaction is :

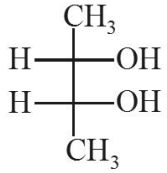


(A) 

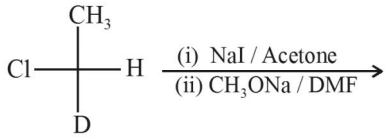
+ it's enantiomer

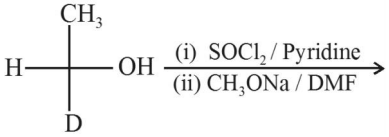
(B) 

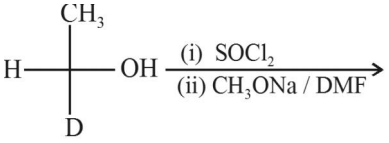
(C) 

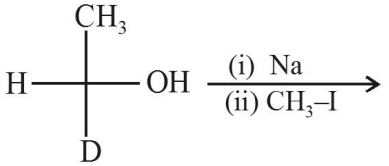
(D) 

7. Which of following reaction sequence shows overall inversion of configuration in product ?

(A) 

(B) 

(C) 

(D) 

8. The reaction for which $K_C \leq K_p$ at 298 K

(A) $3O_2(g) \rightleftharpoons 2O_3(g)$

(B) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(C) $N_2(g) + 2NO_2(g) \rightleftharpoons 2NO_2(g)$

(D) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

9. Which of the following order is NOT correct, against mentioned properties?

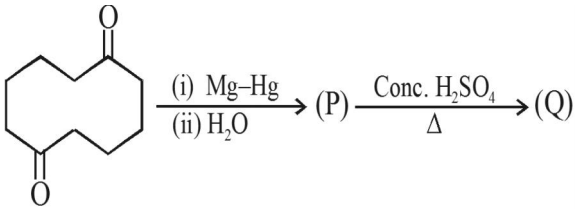
(A) Sc > Y > La (Ionisation energy)

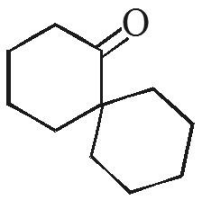
(B) $Hf^{2+} < Hf \simeq Zr$ (size)

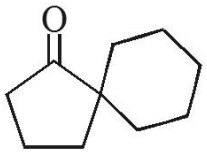
(C) Aℓ > Si (Electronegativity)

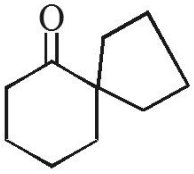
(D) Ga > Aℓ (Ionisation energy)

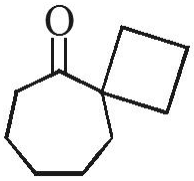
10. Major product (Q) of following reaction is :



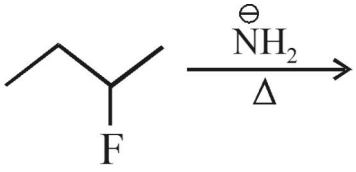
(A) 

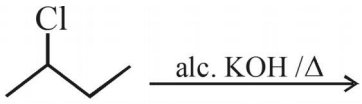
(B) 

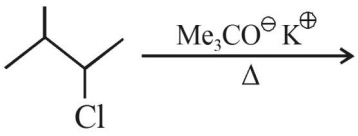
(C) 

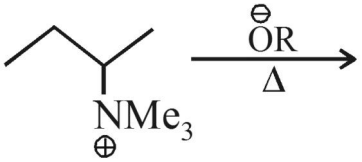
(D) 

11. Which of following reaction(s) produce Saytzeff product as a major product :

(A) 

(B) 

(C) 

(D) 

12. Which of the following liquid pair does not show contraction in volume on mixing -

(A) $CHCl_3+CH_3COCH_3$ (B) $H_2O + HCl$

(C) $H_2O + HNO_3$ (D) $H_2O + C_2H_5OH$

13. In which of the following process energy is absorbed?

(A) $S^{2-}_{(g)} \rightarrow S^{-}_{(g)} + e^{-}$

(B) $N^{-}_{(g)} \rightarrow N_{(g)} + e^{-}$

(C) $O^{-}_{(g)} + e^{-} \rightarrow O^{2-}_{(g)}$

(D) $Na^{+}_{(g)} + e^{-} \rightarrow Na_{(g)}$

14. Consider the following reactions :-

(i) $F_2 + 2X^- \longrightarrow 2F^- + X_2$ (X = Cl, Br, I)

(ii) $Cl_2 + 2X^- \longrightarrow 2Cl^- + X_2$ (X = Br, I)

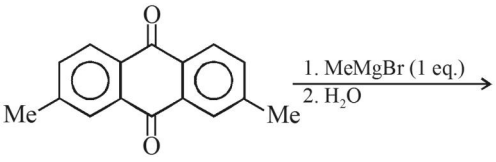
(iii) $I_2 + 2Br^- \longrightarrow Br_2 + 2I^-$

option which have only the correct set of reaction is.

(A) (ii), (iii) (B) (i), (iii)

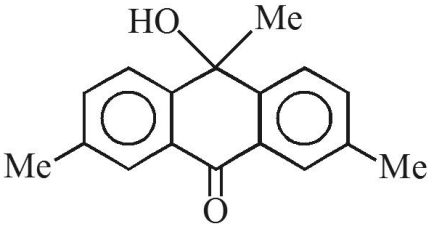
(C) (i), (ii), (iii) (D) (i), (ii)

15.

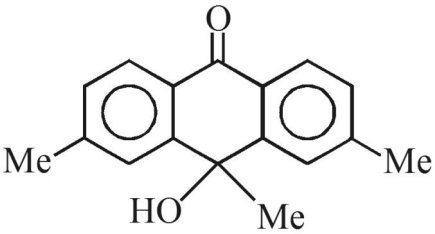


In above reaction out of the following which will be obtained in maximum yield.

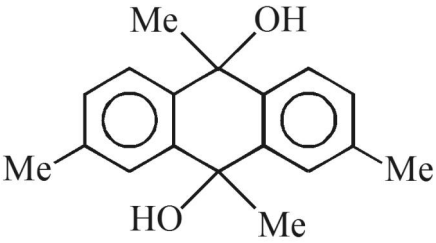
(A)



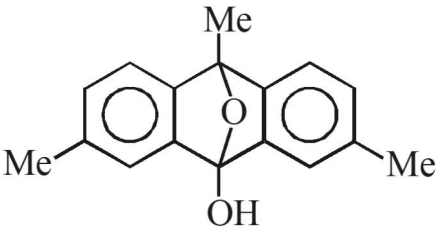
(B)



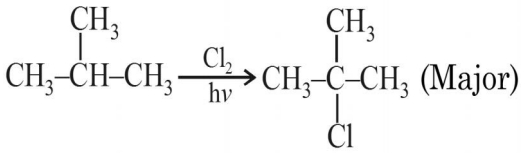
(C)

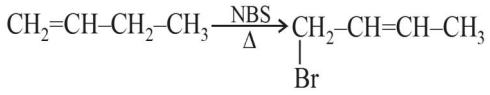


(D)

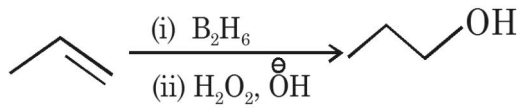


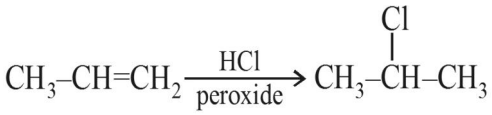
16. Select incorrect option :

(A) 

(B) 

(Major)

(C) 

(D) 

(Major)

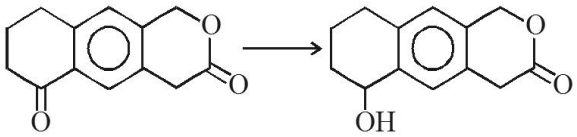
17. Which of the following statement is INCORRECT.

(A) In Haber's process iron oxide, potassium oxide and aluminium oxide are used as catalyst.

(B) ClO_2 is used as bleaching agent

(C) Decomposion of ozone is an endothermic process

(D) Hydrolysis of XeF_6 is a non redox process

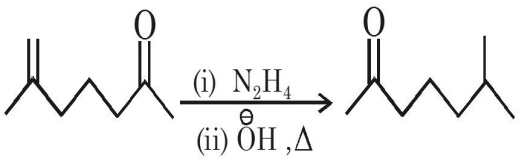
18. 

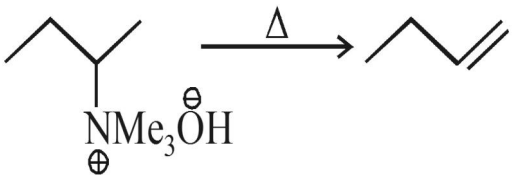
Select best reagent to carry out above conversion.

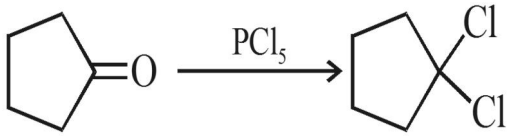
(A) $LiAlH_4$ (B) $NaBH_4$

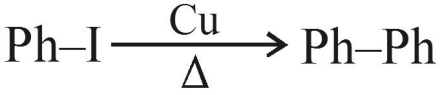
(C) $NH_2–NH_2+H_2O_2$ (D) H_2 (excess) / Ni, Δ

19. Select option in which incorrect major product is shown.

(A) 

(B) 

(C) 

(D) 

20. Which of the following decolorises Br_2 water solution :

(A) 

(B) $CH_3–CH_2–OH$

(C) $HC \equiv CH$

(D) $CH_3–CH_2–Cl$

SEC-B : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

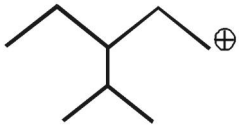
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1. Phosphoric acid is prepared commercially from phosphate ore as following :

$Ca_3(PO4)_2 + SiO_2 + C + O_2 + H_2O \longrightarrow CaSiO_3 + CO_2 + H_3PO_4$

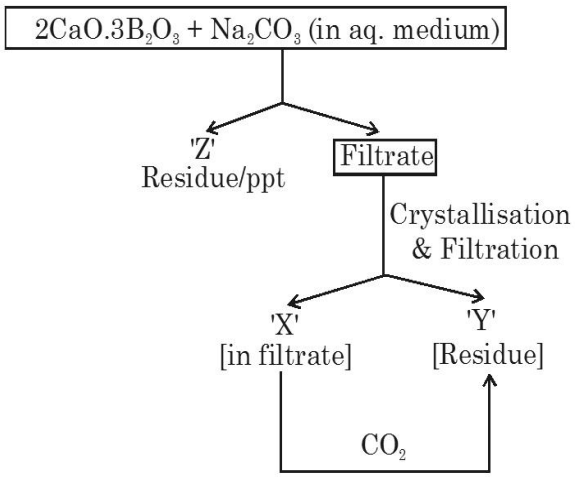
If 1.0 kg each of calcium phosphate and silica are used with excess of C, O_2 and H_2O, what is the maximum quantity of phosphoric acid (in kg) that can be produced. (Ca = 40, P = 31, Si = 28)

2. How many 1,2-shift(s) take(s) place in the following carbocation.



3. The solubility of silver chromate is 1.992×10^{-2} gm /litre. The molar mass of Ag_2CrO_4 is 332 gm/mol. If the molar concentration of Ag^+ in saturated solution of Ag_2CrO_4 is $X \times 10^{-4}$ M, give 'X'.

4. Consider the following flow diagram :-



Give the total sum of Boron atoms present in X and Y respectively.

5. A fixed amount of glucose is dissolved in 100 gm water to form a solution that freezes at –0.2°C. The solution is cooled down to –0.25°C. Find how many grams of ice would have separated.

6. Give the sum of group number and period of element "Uub".

7. The Henry's law constant for solubility of N_2 gas in water at 25°C is 1×10^5 atm. The mole fraction of N_2 in air is 0.8. The number of moles of N_2 from air that will dissolve in 10 moles of H_2O at 25°C and at 5 atm pressure is $X \times 10^{-4}$. Give 'X'-

8. 200 ml of a mixture of 50% H_2, 40% CH_4 and 10% CO_2 is exploded with excess of oxygen in an eudiometer tube. The resultant mixture is cooled to room temperature and is passed through aq. KOH. Find volume contraction observed (in ml) after passing through KOH.

9. 100 ml, 1M H_2SO_4 solution of density 1.5 gm/ml is mixed with 400 ml of water , resulting in a diluted solution of density 1.25 gm/ml. Find molarity of final diluted solution. (Density of water = 1 gm/ml).

10. If the number of S–S, S–O–S and O–H linkages in a molecule of pyrosulphurous acid are x, y and z respectively.
Give (x + y + z).

SEC-A : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : –1 If wrong option is selected.

1. Set of values of 'α' for which the angle bisector of the lines $(\alpha + 1)x + 2y + 5 = 0$ and $4x + \alpha y - 3 = 0$ containing origin is also the obtuse angle bisector, is-

(A) $(-\infty, -\frac{2}{3})$ (B) $(-\frac{2}{3}, \infty)$

(C) $(-\infty, -\frac{2}{3}) \cup (1, \infty)$ (D) $(-1, \infty)$

2. Let mirror image of point A(1,4) in line y = x is point B; mirror image of point B in line y = –x is C and mirror image of C in x-axis is D, then area of triangle ABD is-

(A) 3 sq. units (B) $\frac{3}{2}$ sq. units

(C) 6 sq. units (D) 0 sq. units

3. Let P be a variable point on the ellipse $x^2 + 3y^2 = 3$, then the maximum perpendicular distance of P from the line x – y = 10 is -

(A) $3\sqrt{2}$ (B) $4\sqrt{2}$

(C) $6\sqrt{2}$ (D) $5\sqrt{2}$

4. Two parabolas $x = cy^2 + ay + b$ and $x = cy - y^2$ touch each other at the point (0,1), then the value of 2a + b – c is-

(A) –1 (B) –7 (C) –5 (D) 1

5. The length of common chord of the circles $x^2 + y^2 + 2x + 4y - 20 = 0$ and $x^2 + y^2 + 6x - 8y + 10 = 0$ is-

(A) $5\sqrt{\frac{3}{2}}$ (B) $2\sqrt{\frac{3}{2}}$

(C) 5 (D) $\frac{5\sqrt{5}}{2}$

6. A ray of light travelling along the line x = 2 strikes a parabolic mirror with equation $y^2 - 2y - 4x + 5 = 0$ and gets reflected from its surface then equation of reflected ray may be -

(A) y = 5 (B) x = –1

(C) y = –1 (D) y = 1

7. If for a hyperbola the ratio of length of conjugate Axis to the length of transverse axis is 3 : 2 then the ratio of distance between the focii to the distance between the two directrices is-

(A) 13 : 4 (B) 4 : 13

(C) $\sqrt{13}$: 2 (D) 2 : $\sqrt{13}$

8. If $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors such that $\vec{a}$ is perpendicular to $\vec{b}$ and $\vec{c}$ and $|\vec{a}+\vec{b}+\vec{c}| = 1$, then angle between $\vec{b}$ and $\vec{c}$ is-

(A) π (B) $\frac{\pi}{2}$ (C) 0 (D) $\frac{2\pi}{3}$

9. Minimum distance between two points P and Q on the ellipse $\frac{x^2}{25}+\frac{y^2}{4}=1$, if difference between eccentric angles of P and Q is $\frac{3\pi}{2}$, is-

(A) $2\sqrt{2}$ (B) $2\sqrt{5}$

(C) $\sqrt{29}$ (D) $\sqrt{62}$

10. The value of 'a' for which the three given planes

$P_1 : (a+1)x - y - z = a$

$P_2 : x - (a+1)y - z = a$

$P_3 : x - y - (a+1)z = a$

have no point in common, is-

(A) 0 or –3

(B) –3

(C) 0

(D) No such value of 'a' exists

11. A plane passing through a point (2,2,2) cuts positive semi axes at A,B and C and P(α,β,γ) is the centroid of tetrahedron OABC (where O being origin) then select the correct option -

(A) $\alpha+\beta+\gamma \geq \frac{9}{2}$ (B) $\alpha+\beta+\gamma = 1$

(C) $\alpha\beta\gamma = 2$ (D) $\alpha\beta\gamma \leq 3$

12. If $\vec{p}$ and $\vec{q}$ are unequal unit vectors such that $(\vec{p}-\vec{q}).((2\vec{q}+\vec{p})\times(3\vec{p}-\vec{q})) = |\vec{p}+\vec{q}|$, then angle between $\vec{p}$ and $\vec{q}$ will be-

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$

(C) π (D) 0

13. If $\vec{p}$ and $\vec{q}$ are unit vectors such that $[\vec{p}\ \vec{q}\ \vec{p}\times\vec{q}] = \frac{1}{2}$, then angle between $\vec{p}$ and $\vec{q}$ is -

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$

(C) $\sin^{-1}(\frac{1}{4})$ (D) $\frac{\pi}{2}$

14. If $\vec{x}$ is a unit vector such that $\vec{x}\times(\hat{i}-2\hat{j}+\hat{k}) = -\hat{i}+\hat{k}$, then $\vec{x}$ is -

(A) $-\hat{i}$

(B) $-\frac{1}{3}(2\hat{i}-\hat{j}+2\hat{k})$

(C) $\frac{1}{\sqrt{3}}(\hat{i}-\hat{j}+\hat{k})$

(D) $\frac{1}{3}(2\hat{i}+\hat{j}+2\hat{k})$

15. The distance between directrices of the conic $\sqrt{(x-5)^2+(y-4)^2}+\sqrt{(x-3)^2+(y-2)^2}=6$ is-

(A) $9\sqrt{2}$ (B) $2\sqrt{2}$

(C) 4 (D) $\frac{9}{\sqrt{2}}$

16. Let P is a point on hyperbola $x^2 - y^2 = 4$, which is at minimum distance from (0,–1) then distance of P from x-axis is-

(A) 0 (B) $\frac{1}{2}$

(C) 1 (D) $\sqrt{2}$

17. If image of point P(λ,2λ) in the line mirror $x - y + 2 = 0$ is (μ,3λ), then choose the correct option -

(A) $\mu - \lambda = 0$ (B) $\lambda + \mu = 1$

(C) $\lambda - \mu = -1$ (D) $\lambda + \mu = 0$

18. If angles made by lines $x^2 - 4xy - y^2 = 0$ with positive direction of x-axis are θ_1 and θ_2 then the value of $\sec^2(\theta_1 + \theta_2) + \left|\frac{1}{\tan\theta_1} + \frac{1}{\tan\theta_2}\right|$ is -

(A) 1 (B) 3

(C) 9 (D) 7

19. Equation $x^3 + 8y^3 + 24xy = 64$ represents -

(A) Three straight lines

(B) One straight line and one circle

(C) One straight line and one point

(D) Two straight lines and one point

20. The value of λ for which points A(2,2,1), B(1,1,1), C(–λ,2,1) and D(3,0,–1) are coplanar, is-

(A) $\lambda = -2$ (B) $\lambda = 1$

(C) $\lambda = 2$ (D) $\lambda = -1$

SEC-B : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

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1. Let P(2,3), Q(6,0) and R(α,β) be three points in x-y plane such that |PR + QR| + |PR – QR| is minimum then the value of (α – 2β) is

2. Let A(3,5), B(–2,–7) and C(α,β) are three points such that ∠ACB is a right angle and the area of triangle ABC is $\frac{82}{3}$ square units. Then number of such points C, is-

3. The number of integral values of 'α' for which the abscissa of point of intersection of lines $y = x + 9\alpha$ and $3\alpha x + 2y + 9 = 0$ is integer, is -

4. Let 'C' be the locus of centre of circles which touch the lines $x + 2y - 5 = 0$ and $2x - 4y + 7 = 0$ and area enclosed by curve 'C' with the line $x - y - 5 = 0$ is $\frac{P^2}{2Q^2}$; where P and Q are relative primes, then P + Q is-

5. Let a, b and c are real numbers such that $a^2 + b^2 + c^2 = 1$, then the maximum value of $(4b - 3c)^2 + (4a - 2c)^2 + (3a - 2b)^2$ is -

6. Distance of the point P(1,2,1) from the plane $2x + y - z = 10$ measured along the line $\frac{x-5}{1} = \frac{2y-3}{2} = \frac{z-\frac{5}{2}}{1}$ is -

7. If lines $\frac{x-1}{3} = \frac{y-2}{-1} = \frac{z-\lambda}{2}$ and $\frac{x+1}{-2} = \frac{y}{3\lambda} = \frac{2z-7}{1}$ are coplanar then sum of value(s) of λ is-

8. A plane passes through a fixed point A(1,–2,3). If locus of foot of perpendicular to it from the point B(0,1,2) is $x^2 + y^2 + z^2 + \alpha x + \beta y + \gamma z + \delta = 0$, then the value of $|\alpha + \beta + \gamma + \delta|$ is-

9. Let mirror image of point $A(\alpha,\beta)$ in line mirror $x + 2y = 3$ is point B and image of B in lines $3x - 2y = 5$ is C. If origin is the orthocentre of triangle ABC and P(a,b) is point inside the triangle such that triangles PAB, PBC and PCA have same area then (a + b) is-

10. Let $P(\alpha,\beta)$ be a point on the parabola $y^2 = 4x$ which is at minimum distance from the circle $x^2 + y^2 - 4x - 20y + 103 = 0$, then $\frac{\alpha}{\beta^2}$ is -

Space for Rough Work

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE (Advanced)
MAJOR
04-07-2021

JEE(Main + Advance) : IIT-JEE(M + A) ENTHUSIAST & LEADER (PHASE : ALL PHASES)

Test Type : MAJOR Test Pattern : JEE (Advanced)

TEST DATE : 04-07-2021

ANSWER KEY PAPER-1

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,D	A,C	A,C	B,C	A	B,C	A,C	A,C	B,D	A,B,C
SEC-2	Q.	1	2	3	4	5	6				
	A.	A,C	D	A,C	A,B,C,D	A,C,D	A,B,C				
SEC-3	Q.	1	2								
	A.	A->PQS,B->PRS,C->PQ,D->Q	A->RS,B->QRT,C->RS,D->PS								

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C,D	A,B,C,D	A,B	A,B	A,C,D	A,B,C	A,D	A,B,C	A,B,C,D	B,C,D
SEC-2	Q.	1	2	3	4	5	6				
	A.	C,D	A,C	A,B,C	A,C,D	D	C				
SEC-3	Q.	1	2								
	A.	A->PRS,B->PQT,C->P,D->PQT	A->PQR,B->RT,C->ST,D->S								

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C,D	A,B	A,D	B	B,D	A,B,C,D	A,B	A,B,C	C,D	C
SEC-2	Q.	1	2	3	4	5	6				
	A.	B,C,D	C,D	B	C	A,B	A				
SEC-3	Q.	1	2								
	A.	A->QRT,B->QS,C->QR,D->PT	A->Q,B->P,C->T,D->PQRST								

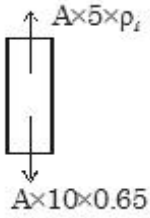
HINT – SHEET

PHYSICS

SEC-1

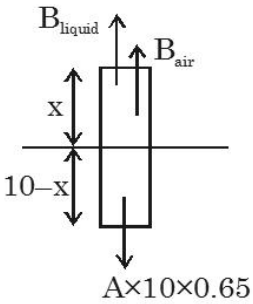
2. Ans (A,C)

Initial



$S_\ell = 1.3$

final



$B_{liquid} = A(10 - x) \times 1.3$

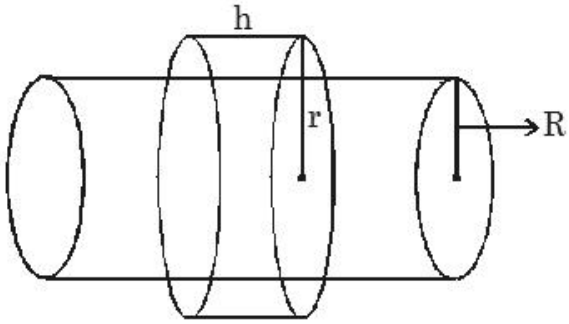
$B_{air} = A \times x \times 0.130.$

$B_{air} + B_{liquid} = mg$

$x = \frac{50}{9}$

Displacement $-5 + x = \frac{50}{9}$

3. Ans (A,C)



$$\frac{\rho \times \pi 2rh}{\varepsilon_0} = E \times 2\pi rh$$

$$E_2 = \frac{3R^2}{2\varepsilon_0 r} = \frac{4\pi G3R^2}{2r} = \frac{2\pi G3R^2}{r}$$

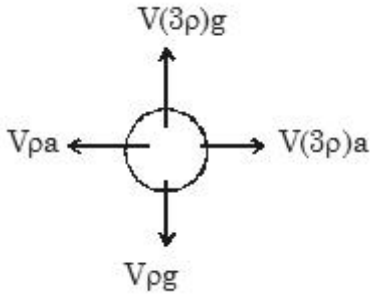
$$E_2 = G \times \frac{\rho \times 4}{3}\frac{\pi R^3}{r^2}$$

$$= \frac{4\pi G\rho R^3}{r}\left(\frac{R}{3r}\right)$$

$$E = G \times \frac{2\pi G\rho R^2}{r}\left(\frac{2R}{3r}\right)$$

$r >> R$

6. Ans (B,C)



Since ratio of displacement along y and x is $\frac{1}{2}$ so there must be same relation between acceleration along y-axis and x-axis.

7. Ans (A,C)

$q = CV$

$i = \frac{dq}{dt} = C\frac{dv}{dt}$

$V_{CD} = iR$

for large value of R & C circuit will behave as filter.

8. Ans (A,C)

Surface of conductor is equipotential so potential at point A and B will be same.

9. Ans (B,D)

mass rate of strike $\frac{dm}{dt} = A(2v)\rho$

Change in velocity during strike

$= \Delta v = \vec{v}_f - \vec{v}_i = (-v) - (v) = -2v$

Thrust force on jet $= \Delta v \frac{dm}{dt} = \frac{dm}{dt}(-2v)$

$= -4\rho Av^2$

Thrust force on wall = – Thrust force on jet = $4\rho Av^2$

$$\frac{d(KE)}{dt} = \frac{1}{2}\frac{dm}{dt}(v_f^2) - \frac{1}{2}\frac{dm}{dt}(v_i^2)$$

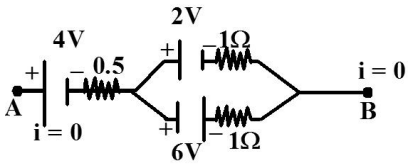
$$= \frac{1}{2}\frac{dm}{dt}\left[(-v)^2 - v^2\right] = 0$$

10. Ans (A,B,C)

Internal current $i = \frac{2+6}{1+1} = 4A$

$V_A - 4 - 2 + 1 \times 4 - V_B = 0$

$V_A - V_B = 2V$



PHYSICS

SEC-2

3. Ans (A,C)

$T = 2\pi\sqrt{\frac{GM}{R^3}}$

$T^2 = \frac{4\pi^2 GM}{R^3}$

$\frac{2\Delta T}{T} = -\frac{3\Delta R}{R}$

$\frac{2 \times 1.08/16}{1.5 \times 3600} = \frac{-3 \times \Delta R}{6400km}$

$\frac{2.16}{24 \times 3600}$

$= \frac{2.16}{86400} \times 6400 \times 10 = 3\Delta R$

$\Delta R = \frac{160}{3} m$

4. Ans (A,B,C,D)

$V = \sqrt{\frac{GM}{r}}$

$\frac{V_A}{V_B} = \sqrt{\frac{r_B}{r_A}} = 2$

$E = \frac{-GMm}{2R}, \frac{E_A}{E_B} = \frac{m_A}{m_B} \times \frac{r_B}{r_A} = 12$

$u_A = \sqrt{\frac{2GM}{r}}, \frac{u_A}{u_B} = 2$

$T = \frac{2\pi}{\sqrt{GM}}(r)^{3/2}$

$\frac{T_A}{T_B} = \left(\frac{r_A}{4\pi A}\right)^{3/2} = \frac{1}{8}$

6. Ans (A,B,C)

$E = -\frac{\partial v}{\partial x}\hat{i} - \frac{\partial v}{\partial y}\hat{j}$

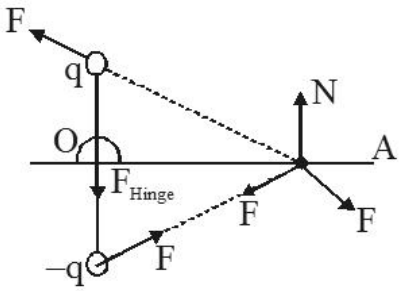
$= -2xy\hat{i} - x^2\hat{j}$

PHYSICS

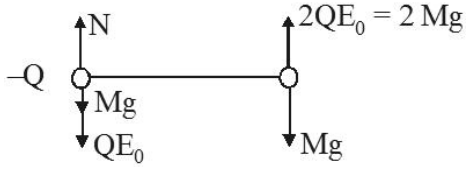
SEC-3

1. Ans (A->PQS,B->PRS,C->PQ,D->Q)

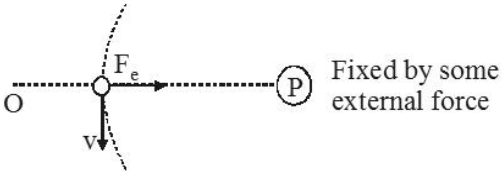
(P)



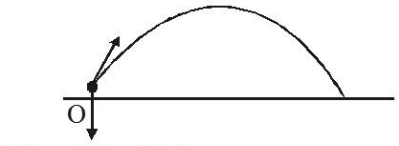
(Q)



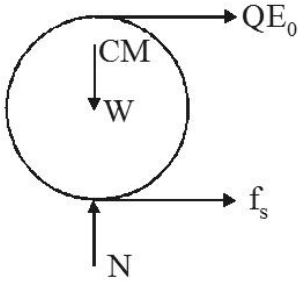
(R)



(S)



(T)



2. Ans (A->RS,B->QRT,C->RS,D->PS)

$V_{eq} = \frac{\frac{1.5}{2} + \frac{V}{1}}{\frac{1}{2} + \frac{1}{1}}$

$= \frac{\left(V + \frac{3}{4}\right)}{\frac{3}{2}} = \frac{2V}{3} + \frac{1}{2}$

$i = \frac{\frac{2V}{3} + \frac{1}{2}}{\frac{2}{3} + 4}$

CHEMISTRY

SEC-1

1. Ans (B,C,D)

$$\text{Moles of CaO} = \frac{4.48}{56} = 0.08 = \text{moles of } CO_2$$

$$\text{Moles of } CaCO_3 = 0.08$$

3. Ans (A,B)

$$XA \longrightarrow Ax(s)$$

$t = 0 \quad 8$

$t = \infty \quad 0 \quad \frac{8}{x}$

$$300 = x_B \times 420 = \frac{5}{\left(5+\frac{8}{x}\right)} \times 420$$

$x = 4$

$$\text{At t = 1hr : } 4A \longrightarrow A_4(\text{solid})$$

$$(8-\alpha) \quad \frac{\alpha}{4}$$

$$\text{Total moles} = \left(8-\alpha+\frac{\alpha}{4}+5\right)$$

$$= (13 - 0.75\,\alpha)$$

$$360 = \left(\frac{8-\alpha}{13-0.75\alpha}\right)\times 340 + \left(\frac{5}{13-0.75\alpha}\right) \times 420$$

$\alpha = 2$

4. Ans (A,B)

Correct order of :-

Electron Affinity : Ne < Aℓ

Ionisation Energy : Li < B

5. Ans (A,C,D)

During hydrolysis of KHF_2, H_2 obtained at cathode and F_2 at anode.

6. Ans (A,B,C)

P_4, S_8, Cl_2, Br_2, I_2 disproportionate in alkali medium.

CHEMISTRY

SEC-2

2. Ans (A,C)

$$9.6 = 9 + \log\left[\frac{\text{Ionized}}{\text{Unionized}}\right]$$

$$\log\left(\frac{\text{Ionized}}{\text{Unionized}}\right) = 0.6 = \log 4$$

% ionized = 80%

% unionized = 20%

3. Ans (A,B,C)

(i) Malonic acid + $P_4O_{10} \longrightarrow C_3O_2 + (HPO_3)_n$

$C_3O_2 \xrightarrow{\Delta} CO_2 + C(s)$ (gas)

(ii) $C(s) \xrightarrow[\text{High temperature}]{SiO_2} Si(s) + CO$ (burns with blue flame)

CO_2 & SO_2 both are acidic, so not distinguish by KOH.

4. Ans (A,C,D)

Oxygen is most abundant in earth crust.

MATHEMATICS

SEC-1

1. Ans (A,C,D)

$am^2 + 2m + 1 = 0$

$m^2 + 2m + a = 0$

Solving we get $m = \pm 1$.

If $m = 1$ is common root then $a = -3$.

Hence other lines are $y = -\frac{x}{3}$ and $y = -3x$.

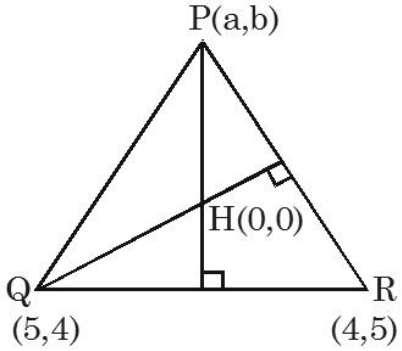
Hence $(x + 3y)(3x + y) = 0$

$\Rightarrow \quad 3x^2 + 10xy + 3y^2 = 0$

$\alpha = 3, \beta = 10, \gamma = 3$.

2. Ans (A,B)

$M_{PH} \times M_{QR} = -1$



$\left(\frac{b}{a}\right)(-1) = -1$

$\Rightarrow a = b \quad ...(1)$

$M_{QH} \times M_{PR} = -1$

$\frac{4}{5} \times \frac{b-5}{a-4} = -1$

$\frac{b-5}{a-4} = -\frac{4}{5} \Rightarrow 4b - 20 = 20 - 5a$

$5a + 4b = 40 \quad ...(2)$

$(a, b) = \left(\frac{40}{9}, \frac{40}{9}\right)$

$[a + b] = \left[\frac{80}{9}\right] = 8$

3. Ans (A,D)

Unit vector along line L_1

$\hat{n}_1 = \frac{2\hat{i} - 2\hat{j} - \hat{k}}{3}; \ \hat{n}_2 = \frac{8\hat{i} + \hat{j} - 4\hat{k}}{9}$

$\hat{n}_a = \hat{n}_1 + \hat{n}_2 = \frac{14\hat{i} - 5\hat{j} - 7\hat{k}}{9}$

$\hat{n}_b = \hat{n}_1 - \hat{n}_2 = \frac{-2\hat{i} - 7\hat{j} + \hat{k}}{9}$

$\vec{r}.\hat{n}_a = 0$ and $\vec{r}.\hat{n}_b = 0$

4. Ans (B)

The general point on line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ is $(\lambda, 2\lambda, 3\lambda)$.

Solve with given plane $x + y + z = 3$ we get $\lambda = \frac{1}{2}$.

So point of intersection A with plane is point of intersection with line

$\frac{x-1}{2} = \frac{y-2}{-1} = \frac{z-3}{-1}$ is B(1, 2, 3)

$\ell_{AB} = \sqrt{\frac{1}{4} + 1^2 + \frac{9}{4}} = \frac{\sqrt{14}}{2} = k$

$\therefore \quad 2k = \sqrt{14}$

5. Ans (B,D)

$\vec{p}.\vec{q} = (x^2\hat{i} - 3\hat{j} + (x+3)\hat{k}).\left(\hat{i} + 3\hat{j} - (x-3)\hat{k}\right)$

$= x^2 - 9 - (x^2 - 9) = 0$

$\Rightarrow \vec{p} \perp \vec{q}$ and $|\vec{p}| = |\vec{q}|$

$\vec{r}.\vec{s} = (2\vec{p} + 3\vec{q}).(3\vec{p} - 2\vec{q}) = 0$

$\therefore \vec{r} \perp \vec{s}$

i.e. $\vec{r} \wedge \vec{s} = \frac{\pi}{2}$

6. Ans (A,B,C,D)

The points of intersection will be

$A(\sqrt{5}, 1)$ & $b(-\sqrt{5}, -1)$.

Suppose the 3rd vertex is $C(2\sec\theta, 2\tan\theta)$, then we know that the orthocentre would also lie on the same rectangular hyperbola since Δ is equilateral so orthocentre coincides with centroid which is $\left(\frac{2\sec\theta}{3}, \frac{2\tan\theta}{3}\right)$.

Which clearly doesn't lie on hyperbola $x^2 - y^2 = 4$. Hence no such equailateral triangle can be formed.

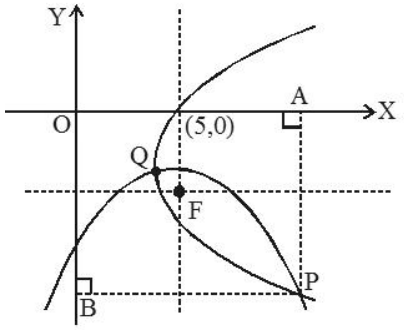
7. Ans (A,B)

The area will be maximum when the line $x + y + 2 = 0$ will be diameter of the given circle.

$\alpha^2 + \beta^2 = 4$

$\therefore \quad (a, b) \equiv (2, 0), (-2, 0), (0, 2), (0, -2)$

8. **Ans (A,B,C)**



Let F be focus.

Now PA = PF = PB

So PQ is the common chord & hence m = –1

9. **Ans (C,D)**

$$[\vec{n}_1\vec{n}_2\vec{n}_3] = 0 \Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2a & 1 \\ a & a^2 & 1 \end{vmatrix} = 0$$

$a = \frac{1}{2};\ 1$

10. **Ans (C)**

$\vec{a}.\vec{b} = 3$

$|\vec{a}|^2|\vec{b}|^2\cos^2\theta = 9$

$\Rightarrow \cos^2\theta = \frac{9}{5 + 4\sin 2\alpha} \leqslant 1$

$\therefore \alpha = \frac{\pi}{4}$ and $\theta = 0$

hence $\vec{a}$ and $\vec{b}$ area collinear i.e.

$$\frac{2(\sin\alpha + \cos\alpha)}{\sin\beta} = \frac{1}{\cos\beta} \Rightarrow \tan\beta = 2\sqrt{2}$$

MATHEMATICS

SEC-2

1. **Ans (B,C,D)**

Solve curves we get

$$y^2 = \frac{144 - 9r^2}{7}$$

$$x^2 = \frac{16r^2 - 144}{7}$$

The equation of tangent to ellipse is

$y = mx \pm \sqrt{16m^2 + 9}$

which is also tangent to given circle so

$r^2m^2 + r^2 = 16m^2 + 9$ .....(i)

Also if A'B'C'D' is square then m = ±1

$\therefore r = \frac{5}{\sqrt{2}}$

2. **Ans (C,D)**

If A'B'C'D' is a square, then tangents $y = \pm x \pm 5$ for which diagonal length A'B' = 0. Area of circumcircle of $\Delta A'B'C'$ is 25π.

Area of circle $C_1 = \frac{25\pi}{2}$

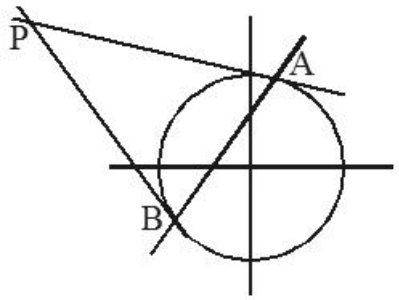
$\therefore$ Required ratio = $\frac{1}{2}$

3. Ans (B)

$C_1 : x^2 + y^2 = 4$

$C_2 : y = x + 1$

while region R is the disc centered at $(\sqrt{3},\ 1)$ and radius



Let $P(x_1,y_1)$

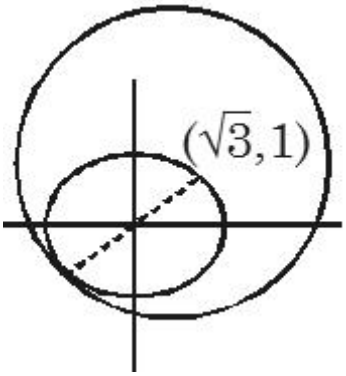
Clearly AB : $y - x = 1$ .....(1)

is chord of contant

$\therefore$ AB : $xx_1 + yy_1 = 4$ .....(2)

(1) & (2) are identical

$\frac{x}{-1} = \frac{y_1}{1} = 4 \Rightarrow \quad (x_1y_1) = (-4, 4)$



4. Ans (C)

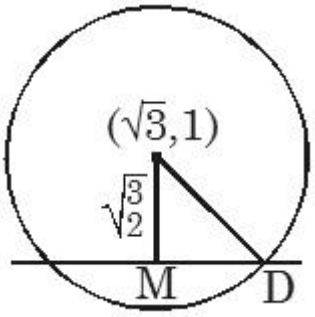
Smallest possible value of 'a' for which

$A \cap C = A$ is $a = 4$.

$\therefore$ R : $(x - \sqrt{3})^2 + (y - 1)^2 \leqslant 4^2$

CD : $y = x + 1$

$P = \left|\frac{\sqrt{3} + 1 - 1}{\sqrt{2}}\right| = \sqrt{\frac{3}{2}}$



$MD = \sqrt{16 - \frac{3}{2}} = \sqrt{\frac{29}{2}}$

$\therefore$ CD $= \sqrt{58}$

5. Ans (A,B)

Equation of family of planes containing the line of intersection of P_1 and P_2 is given by $P_1 + \lambda P_2 = 0$

$\Rightarrow (1 + \lambda)x + (1 - 2\lambda)y + (2 + \lambda)z - 3 - 4\lambda = 0$ .....(1)

Projection of length of line segment AB is greatest, then AB must be parallel to plane (1)

$\therefore$ drs of AB = 2, –1, 3.

$2(1 + \lambda) - 1(1 - 2\lambda) + 3(2 + \lambda) = 0$

$\therefore$ $P_3 : 3y + z + 1 = 0$.

6. Ans (A)

Projection of length of line segment is least is if AB is perpendicular to plane (1).

$\frac{1+\lambda}{2} = \frac{1-2\lambda}{-1} = \frac{2+\lambda}{3} \Rightarrow \ \lambda = 1$

$\therefore$ Equation of plane P_4 is $2x - y + 3z = 7$.

MATHEMATICS

SEC-3

1. Ans (A->QRT,B->QS,C->QR,D->PT)

If $\frac{\beta}{2} = \frac{6}{3} = \frac{8}{4}$

$\Rightarrow \beta = 4$, then lines are parallel.

If $\beta = 4$ and the point on $L_2(3, 5, 5)$ satisfy L_1, then the lines area identical otherwise lines are not identical.

$\therefore \frac{3-1}{2} = \frac{5-2}{3} = \frac{5-\alpha}{4} \Rightarrow \alpha = 1$

If $\beta \neq 4$, then lines are not parallel.

$$D = \begin{vmatrix} \beta & 6 & 8 \\ 2 & 3 & 4 \\ 2 & 3 & 5-\alpha \end{vmatrix}$$

$D = (1-\alpha)(3\beta - 12) = 3(\alpha - 1)(\beta - 4)$.

If $D \neq 0$ lines are skew $\Rightarrow$ $\alpha \neq 1$.

If $D = 0$ lines area intersecting $\Rightarrow \alpha = 1$

2. Ans (A->Q,B->P,C->T,D->PQRST)

(A) $E : \frac{x^2}{2} + \frac{y^2}{3/2} = 1$

$b^2 = 2a^2 + \frac{3}{2}$

$2a^2 - b^2 = -\frac{3}{2} \Rightarrow \frac{b^2}{3/2} - \frac{a^2}{3/4} = 1$

$\frac{y^2}{3/2} - \frac{x^2}{3/4} = 1$

$\therefore e = \sqrt{\frac{3}{2}}$

(B)

$\vec{a} \times \vec{b} = \vec{a} \times (\vec{a} \times \vec{c}) = (\vec{a}.\vec{c})\vec{a} - \vec{a}^2\vec{c} = 2\vec{a} - 3\vec{c}$

$= 2\vec{a} - 3\vec{c}$

$|\vec{a} \times \vec{b} + 3\vec{c} + 2\vec{a}| = |4\vec{a}| = 4\sqrt{3}$

(C) Equation of AB is $S_1 - S_2 = 0$

$-10x + 8y = -14$

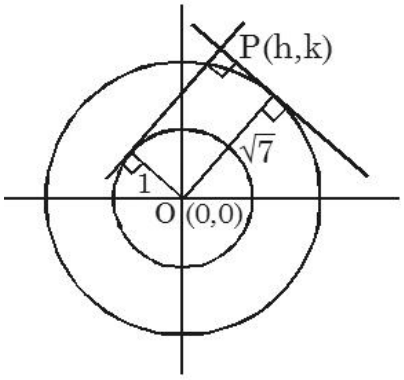
$10x - 8y - 14 = 0$

$5x - 4y - 7 = 0$

$\therefore$ Slope $= \frac{5}{4}$

(D) $h^2 + k^2 = 1 + 7 = 8$

Locus of point P is $x^2 + y^2 = 8$



ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE (Main)
MAJOR
04-07-2021

JEE(Main) : IIT-JEE(M + A) ENTHUSIAST & LEADER (PHASE : ALL PHASES)

Test Type : MAJOR Test Pattern : JEE (Main)

TEST DATE : 04-07-2021

ANSWER KEY

PHYSICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	A	A	B	D	D	C	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	B	A	D	C	A	D	D	C
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	4.00	2.00	9.00	6.00	9.00	2.00	3.00	4.00	2.00	3.00

CHEMISTRY

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	B	C	D	A	C	D	C	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	D	C	D	B	A	C	B	A	C
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	0.63	4.00	1.20	5.00	20.00	19.00	4.00	100.00	0.23	3.00

MATHEMATICS

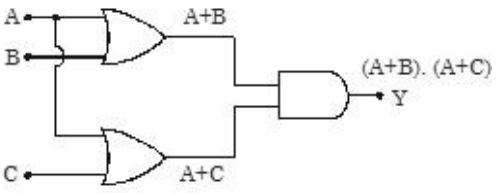
SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	C	C	A	C	A	A	A	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	B	B	A	B	B	C	C	A
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1.00	4.00	1.00	59.00	29.00	6.06	5.05, 5.06	1.00	1.66, 1.67	0.25

HINT – SHEET

PHYSICS

SEC-A

1. Ans (C)



2. Ans (B)

Radio horizon = $\sqrt{2hR_T}$

3. Ans (A)

$i = neAV_{drift} = \frac{V}{R}$

$R = \frac{\rho\ell}{A} = \frac{4\rho\ell}{\pi d^2}$

$\therefore V_{drift} = \frac{V}{RneA} = \frac{V}{\frac{\rho\ell}{A} \times neA} = \frac{V}{\rho\ell ne} \propto \frac{V}{\ell}$

4. Ans (A)

$2k\lambda\ell n\left(\frac{a}{r_1}\right)+2k\lambda\ell n\left(\frac{a}{r_2}\right)+$

$4k\lambda\ell n\left(\frac{a}{r_3}\right)=c$

$=\frac{r_3^2}{r_1r_2}=\text{const.}$

5. Ans (B)

Electric field is in the direction of high potential to low potential. Potential energy = q × potential.

7. Ans (D)

$h_b > h_d > h_c > h_a$. The liquid level is higher where the pressure is lower. The pressure is lower where the flow speed is higher. The flow speed is highest in the narrowest tube, zero in the open air.

8. Ans (C)

Force exerted by oil on the cylinder is zero.

Force by water = $(4\sigma g + 4\rho g) \times A$

$=\left[4\times\frac{1}{2}\times\frac{10^4}{100}+\frac{4\times10^3\times10}{100}\right]\times\frac{16\pi}{10^4}=0.96\pi N$

10. Ans (B)

$T'=2\pi\sqrt{\frac{\ell}{g'}}=\sqrt{2}.\ 2\pi\sqrt{\frac{\ell}{g}}$

$g'=\frac{g}{2}$

$\frac{gR^2}{(R+h)^2}=\frac{g}{2}$

$h^2+2hR-R^2=0$

$h=(\sqrt{2}-1)R$

12. Ans (B)

$a_x=\frac{qE}{m}=10m/s^2$

$\therefore T=\frac{2\times10}{10}=2\text{ sec}$

$R=10\times2-\frac{1}{2}\times10\times4=0$

$\because a_x=a_y \Rightarrow$ particle will follow a straight line.

15. Ans (D)

By Kepler's law

$T^2=\frac{4\pi^2}{GM}R^3=\frac{4\pi^2}{G\rho}\times\frac{4\pi}{3}$

R = R asteroid

$\Rightarrow$ so, time period depends only on density of planet

T = TP of near earth orbit satellite = 84 min = 1 hr 24 min

16. Ans (C)

We know that

$\beta=\frac{\alpha}{1-\alpha}=\frac{0.96}{1-0.96}=24$

The collector current I_C is given by

$I_C=\frac{V_C}{R}=\frac{0.5}{800}=0.625\times10^{-3}A$

Further $I_B=\frac{I_C}{\beta}=\frac{0.625\times10^{-3}}{24}$

$=26\times10^{-6}\text{ A}=26\ \mu A$

17. Ans (A)

$R_L=24\times10^{\circ}\ \Omega$

$R_f=\frac{R_i}{2}=12\times10^{\circ}\Omega$

18. Ans (D)

$\frac{\varepsilon R_1}{R_1+R_2}=\frac{\varepsilon C_2}{C_1+C_2}$

$R_1C_1+R_1C_2=R_1C_2+R_2C_2$

or $R_1C_1=R_2C_2$

19. Ans (D)

$mv_0R=mva$ ....(i)

$\frac{1}{2}mv_0^2=\frac{1}{2}mv^2-\frac{GMm}{a}$ ....(ii)

20. Ans (C)

Truth table for NAND gate

x	y	z
0	0	1
0	1	1
1	0	1
0	0	0

PHYSICS

SEC-B

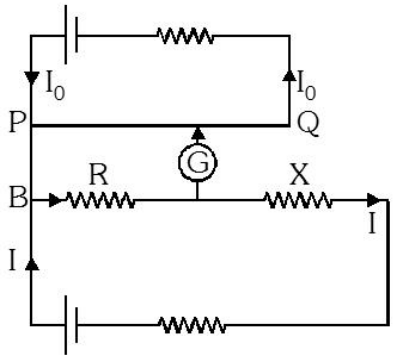
1. Ans (4.00)

$$\frac{1}{2}\left[\frac{q_{en}}{\varepsilon_0}\right] = \phi$$

$$\phi = \frac{4q}{\varepsilon_0}$$

2. Ans (2.00)

In potentiometer wire potential difference is directly proportional to length



Let potential drop unit length a potentiometer wire be K.

For zero deflection the current will flow independently in two circles

$IR = K \times 10$ ...(1)

$IR + IX = K \times 30$ ...(2)

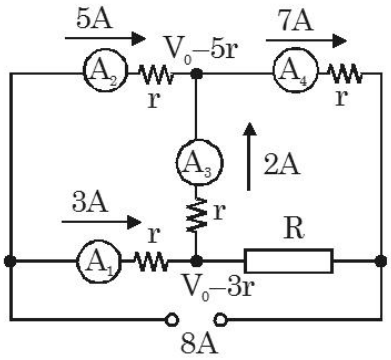
(2) – (1)

$\Rightarrow IX - k \times 20$ ...(3)

Divide (1) and (2)

$\frac{R}{X} = \frac{1}{2} \Rightarrow x = 2R$

3. Ans (9.00)



$V_0 - 5r - 7r$

$= V_0 - 12r$

$V_0 - 3r - 1 \times R = V_0 - 12\,r$

$\Rightarrow R = 9r$

4. Ans (6.00)

$$\frac{x-10}{10} + \frac{x+6}{20} + \frac{x}{10} + \frac{x-10}{5} = 0$$

$\Rightarrow x = 6$ V

$\therefore I_{20\,\Omega} = \frac{6+6}{20} = 0.6$ Amp

5. Ans (9.00)

Energy released = q.v.

q = area under curve $= \frac{1}{2} \times 150 \times 0.2 = \frac{30}{2}$C

$v = E.\,d = 400 \times 10^3 \times 1500 = 60 \times 10^7$

energy released $= 60 \times 10^7 \times 15 = 900 \times 10^7 = 9 \times 10^9$ J

6. Ans (2.00)

$$V = \frac{kQ}{R} = \frac{\sigma \times 2\pi R^2}{4\pi\epsilon_0 R} = \frac{\sigma R}{2\epsilon_0}$$

8. Ans (4.00)

$$P = 2 \times \left(\frac{T}{R_1} + \frac{T}{R_2}\right) = 2 \times \left(\frac{T}{5R} + \frac{T}{R}\right) = \frac{12T}{5R}$$

9. Ans (2.00)

In steady state current in the branch containing the capacitor is zero & hence emf E is shared between

r & R_2 = in the ratio of their resistance voltage across R_2 is $\frac{ER_2}{R_2 + r} = 2$ Volts

= Voltage across capacitor.

$\therefore\ Q = CV_C = 2\mu C.$

10. Ans (3.00)

$$V_C = \frac{4CV - CV}{3C} = V$$

$$\therefore U_f = \frac{1}{2} \times (C + 2C)V^2 = \frac{3}{2}CV^2$$

CHEMISTRY

SEC-A

4. Ans (C)

The reaction is exothermic ($\Delta H < 0$)

slope $= -\frac{\Delta H}{R}$

17. Ans (C)

$XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF$

CHEMISTRY

SEC-B

1. Ans (0.63)

$Ca_3(PO_4)_2 + 3SiO_2 + 5C + 5O_2 + 3H_2O \longrightarrow$
$3CaSiO_3 + 5CO_2 + 2H_3PO_4$

L.R.

4. Ans (5.00)

$2CaO \cdot 3B_2O_3 + 2Na_2CO_3 \longrightarrow 2CaCO_3 + Na_2B_4O_7 + 2NaBO_2$

Colemanite

Filtered – $CaCO_3$ (as residue)

$NaBO_2$ (in filtrate) + $Na_2B_4O_7 \cdot 10H_2O$ (as residue) ← Concentrated and allowed to crystallise out and filtered ← $Na_2B_4O_7 + NaBO_2$ (in solution)

CO_2 passed and crystallise out again $[4NaBO_2 + CO_2 \longrightarrow Na_2B_4O_7 + Na_2CO_3]$

$Na_2B_4O_7 \cdot 10H_2O \downarrow$

$\therefore$ x is $NaBO_2$ & y is $Na_2B_4O_7.10H_2O$

Total number of Boron atoms = 1 + 4 = 5.00

5. Ans (20.00)

Let 'x' mole glucose be dissolved.

$$\left.\begin{aligned} 0.2 &= K_f \times \frac{x}{0.1} \\ 0.25 &= K_f \times \frac{x}{m} \end{aligned}\right\} \; m = 80 \text{ gm}$$

80 gm water is left hence, 20 gm ice will freeze out.

6. Ans (19.00)

112–[Rn], $7s^2$, $5f^{14}$, $6d^{10}$

Group number = 12, Period Number = 7

Answer is 12 + 7 = 19

7. Ans (4.00)

$P_{N_2} = 0.8 \times 5 = 4\text{atm}$

$P_{N_2} = K_H \times x$

$4 = 1 \times 10^5 \times \left(\frac{n_{N_2}}{10}\right) \Rightarrow n_{N_2} = 4 \times 10^{-4}$

8. Ans (100.00)

$CH_4 + 2O_2\,(g) \longrightarrow CO_2(g) + 2H_2O\,(liq.)$

80 ml ↓ 80 ml

Total volume of CO_2 = 80 + 20 = 100 ml

9. Ans (0.23)

Moles of H_2SO_4 in initial sol = 0.1 mole

Mass of initial sol = 150 gm

Mass of water = 400 gm

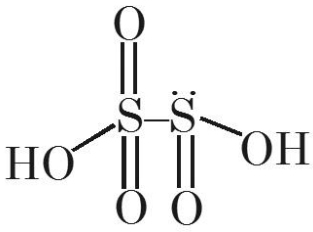
Total mass of sol = 550 gm

Final volume = $\frac{550}{1.25} = 440\text{ml}$

$M = \frac{0.1}{0.44} = 0.227 \simeq 0.23$

10. Ans (3.00)

$H_2S_2O_5$



S–S ⇒ x = 1

S–O–S ⇒ y = 0

O–H ⇒ z = 2

Ans.(1 + 0 + 2 = 3)

MATHEMATICS

SEC-A

1. Ans (A)

$(\alpha + 1)x + 2y + 5 = 0$

$-4x - \alpha y + 3 = 0$

$a_1a_2 + b_1b_2 > 0$

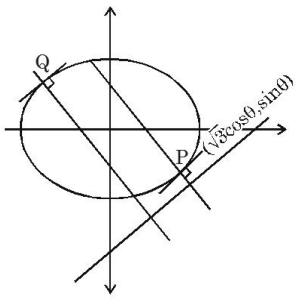
$\Rightarrow -6\alpha > 4 \quad \Rightarrow \alpha < -\frac{2}{3}$

2. Ans (A)

A(1, 4), B(4, 1), C(–1, –4) and D(–1, 4)

Area of ΔABD = 3 sq. units

3. Ans (C)



$$\frac{x^2}{3} + \frac{y^2}{1} = 1$$

$$\frac{dy}{dx} = \frac{-x}{3y}$$

$$= \frac{-\sqrt{3}\cos\theta}{3\sin\theta}$$

$$= \frac{-\cot\theta}{\sqrt{3}}$$

$$\frac{-\cot\theta}{\sqrt{3}} = 1 \Rightarrow \cot\theta = -\sqrt{3}$$

$\Rightarrow \theta = -30°$ or $150°$

For Q, $\theta = 150° \Rightarrow Q\left(\frac{-3}{2}, \frac{1}{2}\right)$

$$\text{Maximum distance} = \left|\frac{\frac{-3}{2} - \frac{1}{2} - 10}{\sqrt{2}}\right| = 6\sqrt{2}$$

4. Ans (C)

$$\left.\begin{matrix} 1 + a + b = 0 \\ \text{and } c - 1 = 0 \end{matrix}\right\} \; c = 1 \text{ and } a + b = -1$$

Now $\frac{dx_1}{dy_1} = 2cy + a = a + 2y$

$$\left.\frac{dy_1}{dx_1}\right]_{(0,1)} = \frac{1}{a+2}$$

and $\frac{dx}{dy} = c - 2y = 1 - 2y$

$$\left.\frac{dy_2}{dx_2}\right]_{(0,1)} = \frac{1}{1-2} = -1$$

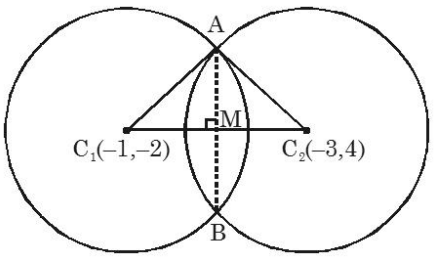
$$\left.\frac{dy_1}{dx_1}\right)_{(0,1)} \left(\frac{dy_2}{dx_2}\right) = -1$$

$$\Rightarrow \left(\frac{1}{a+2}\right)(-1) = -1$$

$\Rightarrow a + 2 = 1 \Rightarrow \boxed{a = -1}$ b = 0

2a + b – c = –2 + 0 + 1 = –1

5. Ans (A)



$r_1 = 5, \quad r_2 = \sqrt{15}$

$c_1c_2 = \sqrt{4 + 36} = \sqrt{40}$

common chord $S_1 - S_2 = 0$

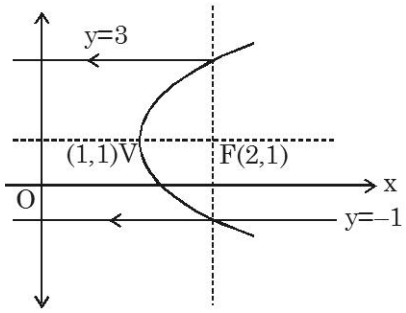
$\Rightarrow \quad 2x - 6y + 15 = 0$

$$C_1M = p = \left|\frac{-2 + 12 + 15}{\sqrt{40}}\right| = \frac{25}{\sqrt{40}}$$

$$AB = 2\sqrt{25 - \frac{625}{40}} = 2\sqrt{\frac{375}{40}}$$

$$= 2\sqrt{\frac{75}{8}} = 5\sqrt{\frac{3}{2}}$$

6. Ans (C)



$(y-1)^2 = 4x-4$

$(y-1)^2 = 4(x-1)$

$y-1 = \pm 2$

Equation of reflected rays

$y = -1$ and $y = 3$

7. Ans (A)

$\frac{b}{a} = \frac{3}{2}$, Now

$\frac{2ae}{\frac{2a}{e}} = e^2 = 1 + \frac{b^2}{a^2} = 1 + \frac{9}{4} = \frac{13}{4}$

8. Ans (A)

$\vec{a}.\vec{b} = 0$ and $\vec{a}.\vec{c} = 0$

$\left|\vec{a}+\vec{b}+\vec{c}\right|^2 = 1+1+1+2\left(\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}\right) = 1$

$\Rightarrow \vec{c}.\vec{a} = -1 \Rightarrow \vec{c} \wedge \vec{a} = \pi$

9. Ans (A)

$P(5\cos\theta, 2\sin\theta)$, $Q(5\sin\theta, -2\cos\theta)$

$PQ = \sqrt{(5\cos\theta - 5\sin\theta)^2 + (2\sin\theta + 2\cos\theta)^2}$

$PQ = \sqrt{29 - 21\sin 2\theta}$

Minmum value of $PQ = \sqrt{8} = 2\sqrt{2}$

10. Ans (B)

$$D = \begin{vmatrix} a+1 & -1 & -1 \\ 1 & -(a+1) & -1 \\ 1 & -1 & -(a+1) \end{vmatrix} = a^2(a+3)$$

$\Rightarrow a^2(a+3) = 0 \Rightarrow a = 0, -3$

four $a = 0$ there are infinite solutions and for $a = -3$, there is no solution.

11. Ans (A)

Let equation of plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

passes through (2, 2, 2)

i.e. $\frac{2}{a} + \frac{2}{b} + \frac{2}{c} = 1 \Rightarrow \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{2}$

$\alpha = \frac{a}{4} \Rightarrow a = 4\alpha$

Similarly $b = 4\beta$ and $c = 4\gamma$

$\Rightarrow \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 2$

For three positive numbers α, β and γ

A.M. $\geq$ H.M.

$\frac{\alpha+\beta+\gamma}{3} \geqslant \frac{3}{\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}} \Rightarrow \alpha+\beta+\gamma \geqslant \frac{9}{2}$

12. Ans (C)

$\vec{p}-\vec{q}$, $2\vec{q}+\vec{p}$ and $3\vec{p}-\vec{q}$ are coplanar

L.H.S. = 0 $\Rightarrow$ R.H.S. = 0 $\Rightarrow \vec{p} = -\vec{q}$

$\vec{p} \wedge \vec{q} = \pi$

13. Ans (B)

$(\vec{p}\times\vec{q}).(\vec{p}\times\vec{q}) = \frac{1}{2}$

$\Rightarrow \sin^2\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{4}$

14. Ans (B)

Let $\vec{x} = a\hat{i} + b\hat{j} + c\hat{k}$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a & b & c \\ 1 & -2 & 1 \end{vmatrix} = (b+2c)\hat{i} - \hat{j}(a-c) + \hat{k}(-2a-b)$$

$= -\hat{i} + \hat{k}$

$b + 2c = -1$, $a - c = 0$ and $2a + b = -1$

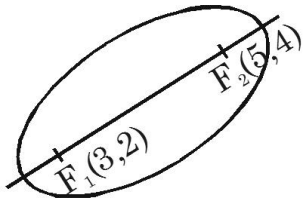
and $a^2 + b^2 + c^2 = 1$

$\Rightarrow a = 0, b = -1, c = 0$

or $a = -\frac{2}{3}, b = \frac{1}{3}, c = \frac{-2}{3}$

$\Rightarrow \vec{x} = -\hat{j}$ or $\vec{x} = -\frac{1}{3}\left(2\hat{i} - \hat{j} + 2\hat{k}\right)$

15. Ans (A)



$2ae = 2\sqrt{2}$

$\Rightarrow ae = \sqrt{2}$

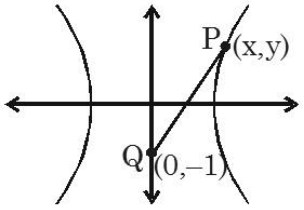
$2a = 6 \Rightarrow a = 3$

$\Rightarrow c = \frac{\sqrt{2}}{3}$

Distance between directrices

$= 2\left(\frac{3}{\frac{\sqrt{2}}{3}}\right) = 9\sqrt{2}$

16. Ans (B)



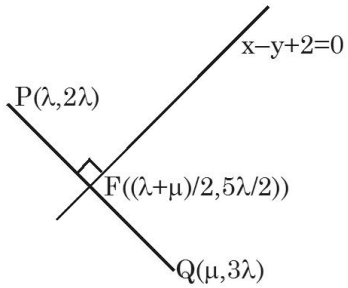
$PQ = \sqrt{x^2 + (y+1)^2}$

$= \sqrt{y^2 + 4 + y^2 + 2y + 1}$

$= \sqrt{2\left(y + \frac{1}{2}\right)^2 + \frac{9}{2}}$

PQ is minimum if $y = -\frac{1}{2}$

17. Ans (B)



$\frac{\lambda + \mu}{2} - \frac{5\lambda}{2} + 2 = 0$

$\mu - 4\lambda + 4 = 0 \quad ...(1)$

and $m_{PQ} = \frac{\lambda}{\mu - \lambda} = -1 \quad ...(2)$

$\Rightarrow \boxed{\mu = 0}$ and $\boxed{\lambda = 1}$

18. Ans (C)

$1 - 4m - m^2 = 0 \Rightarrow m^2 + 4m - 1 = 0$

$m_1 + m_2 = -4$ and $m_1 m_2 = -1$

$\tan(\theta_1 + \theta_2) = \frac{-4}{1+1} = -2$

$\sec^2(\theta_1 + \theta_2) = 5$

and

$\left|\frac{1}{\tan\theta_1} + \frac{1}{\tan\theta_2}\right| = \left|\frac{\tan\theta_1 + \tan\theta_2}{\tan\theta_1 + \tan\theta_2}\right| = 4$

19. Ans (C)

$(x)^3 + (2y)^3 + (-4)^3 = 3(x)(2y)(-4)$

$\Rightarrow x = 2y = -4 \quad \Rightarrow x = -4, y = -2$

i.e. a point (–4, –2)

or $x + 2y - 4 = 0$ i.e. a line

20. Ans (A)

$\left[\overrightarrow{AB}\ \overrightarrow{AC}\ \overrightarrow{AD}\right] = 0$

$\Rightarrow \begin{vmatrix} -1 & -1 & 0 \\ -\lambda - 2 & 0 & 0 \\ 1 & -2 & -2 \end{vmatrix} = 0$

$\Rightarrow \lambda = -2$

MATHEMATICS

SEC-B

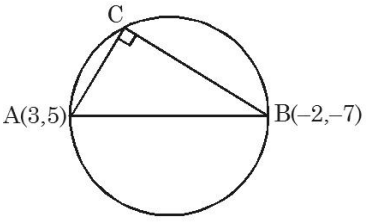
1. Ans (1.00)

R will be the mid point of PQ

i.e. $R\left(4, \frac{3}{2}\right) \Rightarrow \alpha = 4, \beta = \frac{3}{2}$

$\alpha - 2\beta = 4 - 3 = 1$

2. Ans (4.00)



$AB = \sqrt{25 + 144} = 13$

Maximum area of

$\Delta ABC = \frac{1}{2}(13)\left(\frac{13}{2}\right) = \frac{169}{4}$

So 4 such points C are possible.

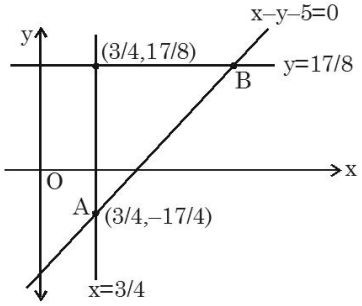
3. Ans (1.00)

$3\alpha x + 2x + 18\alpha + 9 = 0$

$\Rightarrow x = \frac{-18\alpha - 9}{3\alpha + 2} = -6 + \frac{3}{3\alpha + 2}$

Only possible integral value of α is -1.

4. Ans (59.00)



Locus of centre will be angle bisectors of given lines i.e. $x = \frac{3}{4}$ and $y = \frac{17}{8}$

Required area

$= \frac{1}{2}\left(\frac{51}{8}\right)^2 \Rightarrow P = 51,\ Q = 8$

$P + Q = 59$

5. Ans (29.00)

$\vec{r}_1 = a\hat{i} + b\hat{j} + c\hat{k}$

$\vec{r}_2 = 2\hat{i} + 3\hat{j} + 4\hat{k}$

$$|\vec{r}_1 \times \vec{r}_2| = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a & b & c \\ 2 & 3 & 4 \end{vmatrix}$$

$= \left|\hat{i}(4b - 3c) - \hat{j}(4a - 2c) + \hat{k}(3a - 2b)\right|$

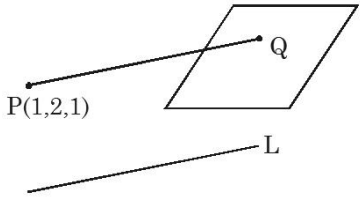
$= \left(\sqrt{a^2 + b^2 + c^2}\right)(\sqrt{29})\sin\theta$

$\Rightarrow$

$(4b - 3c)^2 + (4a - 2c)^2 + (3a - 2b)^2 = 29\sin^2\theta$

Max. value = 29

6. Ans (6.06)



Equation of line PQ

$$\frac{x-1}{1}=\frac{y-2}{1}=\frac{z-1}{1}=\lambda \text{ (say)}$$

$Q(\lambda+1, \lambda+2, \lambda+1)$.

Now Q lies on the plane

i.e. $2(\lambda+1)+(\lambda+2)-(\lambda+1)=10$

$$\Rightarrow 2\lambda=7 \Rightarrow \lambda=\frac{7}{2} \Rightarrow Q\left(\frac{9}{2}, \frac{11}{2}, \frac{9}{2}\right)$$

$$PQ=\sqrt{\frac{49}{4}+\frac{49}{4}+\frac{49}{4}}=\frac{7}{2}\sqrt{3}$$

7. Ans (5.05, 5.06)

$$\begin{vmatrix} 2 & 2 & \lambda-\frac{7}{2} \\ 3 & -1 & 2 \\ -2 & 3\lambda & \frac{1}{2} \end{vmatrix}=0$$

$$\Rightarrow 2\left(-\frac{1}{2}-6\lambda\right)-2\left(\frac{3}{2}+4\right)+\left(\lambda-\frac{7}{2}\right)(9\lambda-2)=0$$

$$\Rightarrow 18\lambda^2-91\lambda-10=0$$

$$\Rightarrow \lambda_1+\lambda_2=\frac{91}{18}=\frac{182}{36}$$

8. Ans (1.00)

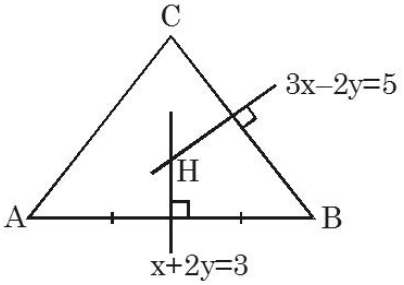
Let foot of $\perp$ be P(x, y, z) then $\overrightarrow{AP}\cdot\overrightarrow{PB}=0$

$\Rightarrow (x-0)(x-1)+(y-1)(y+2)+(z-2)(z-3)=0$

$x^2+y^2+z^2-x+y-5z+4=0$

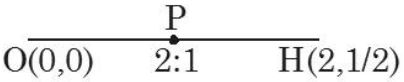
$\alpha+\beta+\gamma+\delta=-1+1-5+4=-1$

9. Ans (1.66, 1.67)



Lines $3x-2y=5$ and $x+2y=3$ are perpendicular bisectors of the triangle ABC

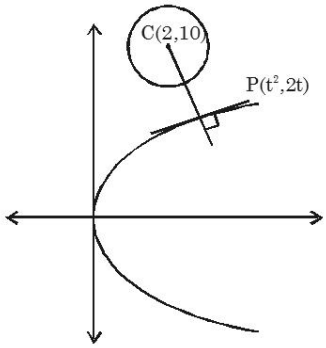
Hence circumcentre H $\left(2, \frac{1}{2}\right)$ and point P is centroid.



$$P\left(\frac{4}{3}, \frac{1}{3}\right) \Rightarrow a+b=\frac{5}{3}$$

10. Ans (0.25)

Radius $=\sqrt{4+100}-103=1$



$$\text{Slope of CP}=\frac{-1}{\left.\frac{dy}{dx}\right]_{at\ 't'}}=-t=\frac{2t-10}{t^2-2}$$

$\Rightarrow -t^3+2t=2t-10 \quad \Rightarrow \quad t=(10)^{1/3}$

$$P\left((10)^{2/3}, 2(10)^{1/3}\right)$$

$$\Rightarrow \alpha=(10)^{2/3}, \ \beta=2(10)^{1/3}$$

$\alpha\beta=2(10)=20$

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE (Main)
MAJOR
04-07-2021

JEE(Main) : IIT-JEE(M + A) ENTHUSIAST & LEADER (PHASE : ALL PHASES)

Test Type : MAJOR Test Pattern : JEE (Main)

TEST DATE : 04-07-2021

ANSWER KEY

PHYSICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	A	A	B	D	D	C	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	B	A	D	C	A	D	D	C
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	4.00	2.00	9.00	6.00	9.00	2.00	3.00	4.00	2.00	3.00

CHEMISTRY

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	B	C	D	A	C	D	C	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	D	C	D	B	A	C	B	A	C
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	0.63	4.00	1.20	5.00	20.00	19.00	4.00	100.00	0.23	3.00

MATHEMATICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	C	C	A	C	A	A	A	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	B	B	A	B	B	C	C	A
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1.00	4.00	1.00	59.00	29.00	6.06	5.05, 5.06	1.00	1.66, 1.67	0.25

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE (Advanced)
MAJOR
04-07-2021

JEE(Main + Advance) : IIT-JEE(M + A) ENTHUSIAST & LEADER (PHASE : ALL PHASES)

Test Type : MAJOR Test Pattern : JEE (Advanced)

TEST DATE : 04-07-2021

ANSWER KEY PAPER-1

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,D	A,C	A,C	B,C	A	B,C	A,C	A,C	B,D	A,B,C
SEC-2	Q.	1	2	3	4	5	6				
	A.	A,C	D	A,C	A,B,C,D	A,C,D	A,B,C				
SEC-3	Q.	1	2								
	A.	A->PQS,B->PRS,C->PQ,D->Q	A->RS,B->QRT,C->RS,D->PS								

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C,D	A,B,C,D	A,B	A,B	A,C,D	A,B,C	A,D	A,B,C	A,B,C,D	B,C,D
SEC-2	Q.	1	2	3	4	5	6				
	A.	C,D	A,C	A,B,C	A,C,D	D	C				
SEC-3	Q.	1	2								
	A.	A->PRS,B->PQT,C->P,D->PQT	A->PQR,B->RT,C->ST,D->S								

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C,D	A,B	A,D	B	B,D	A,B,C,D	A,B	A,B,C	C,D	C
SEC-2	Q.	1	2	3	4	5	6				
	A.	B,C,D	C,D	B	C	A,B	A				
SEC-3	Q.	1	2								
	A.	A->QRT,B->QS,C->QR,D->PT	A->Q,B->P,C->T,D->PQRST								